Resolving a Conjecture on Stack-Sorting

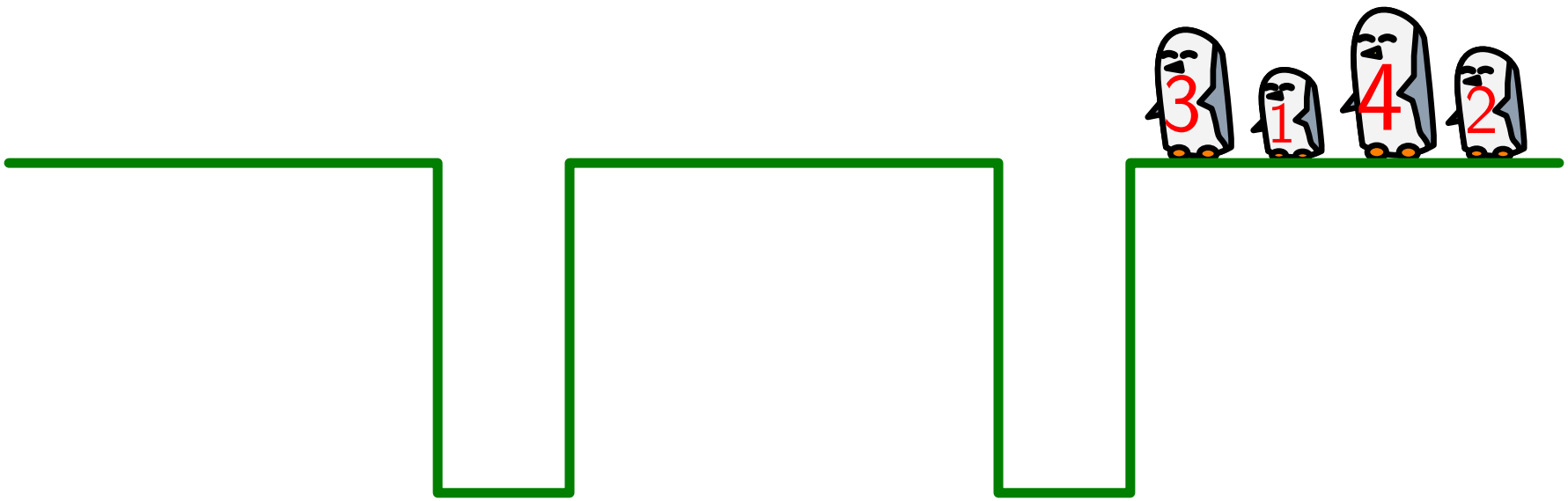
Michael Luo
Research Science Institute
Mentor: Ryota Inagaki

August 2026

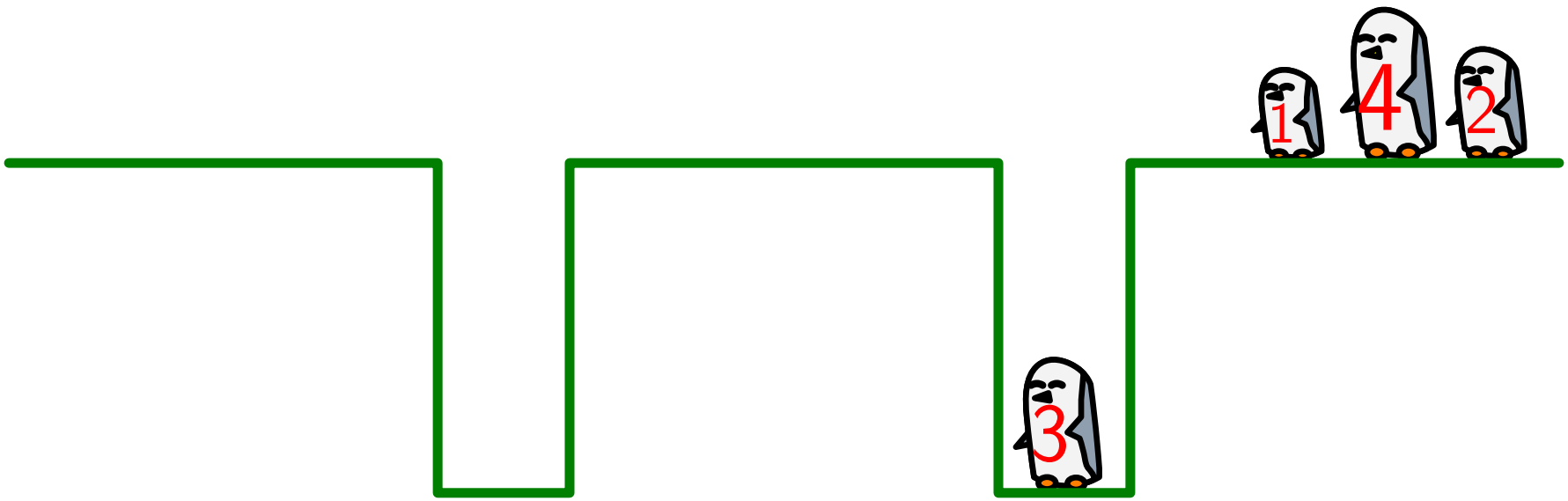
Penguins

Michael Luo
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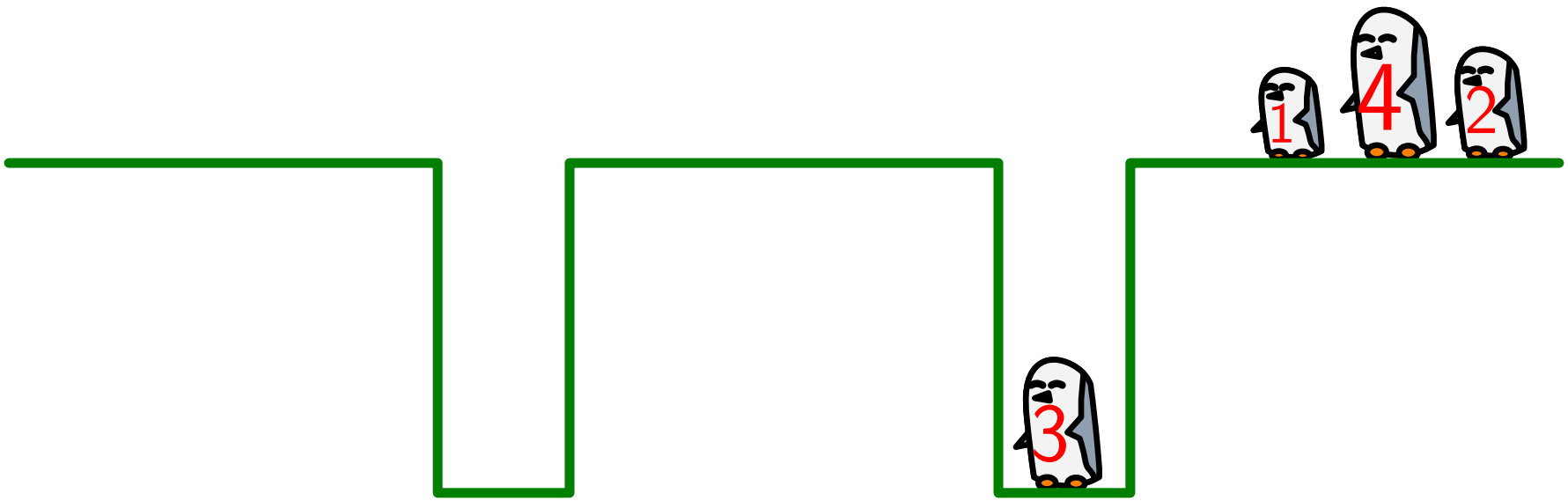
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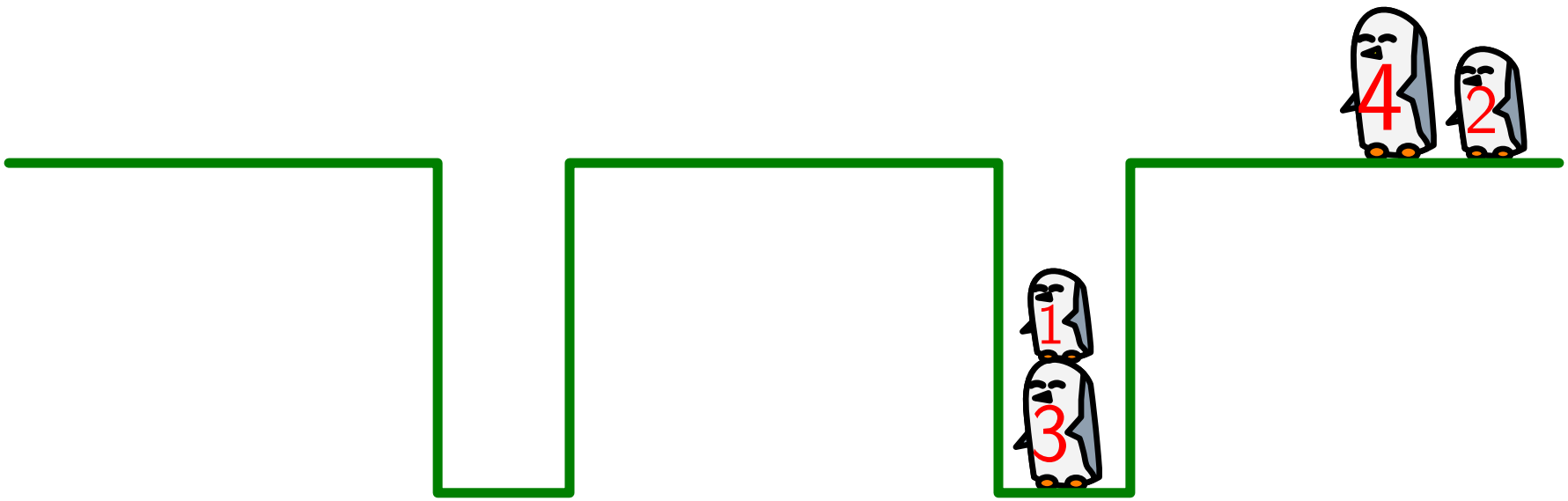
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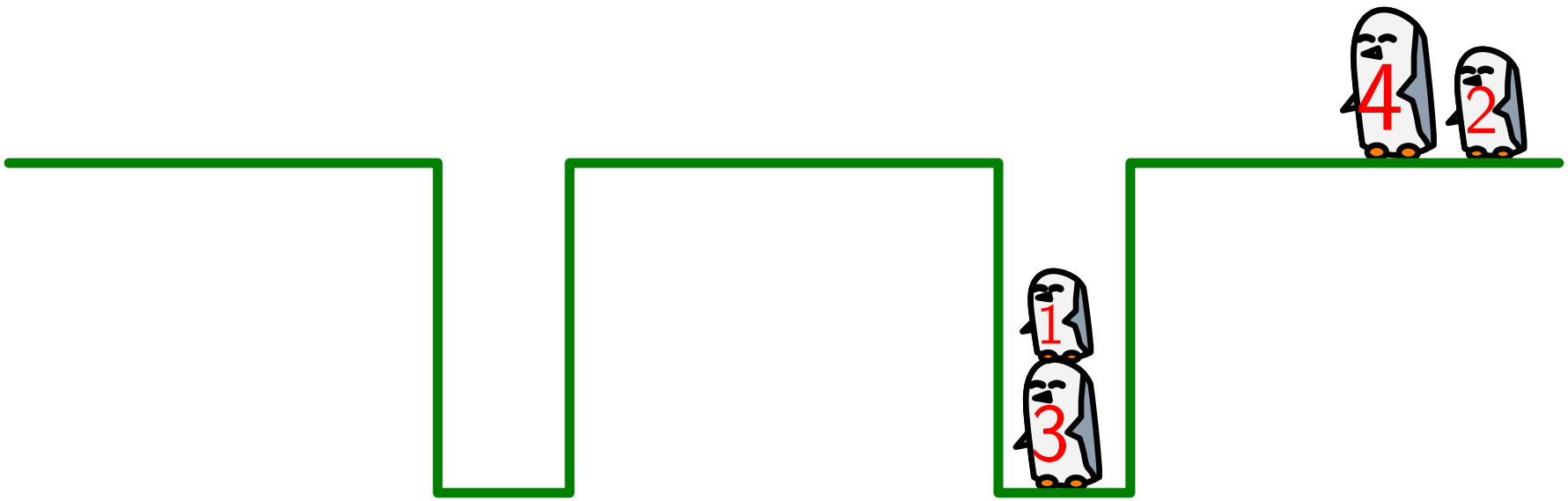
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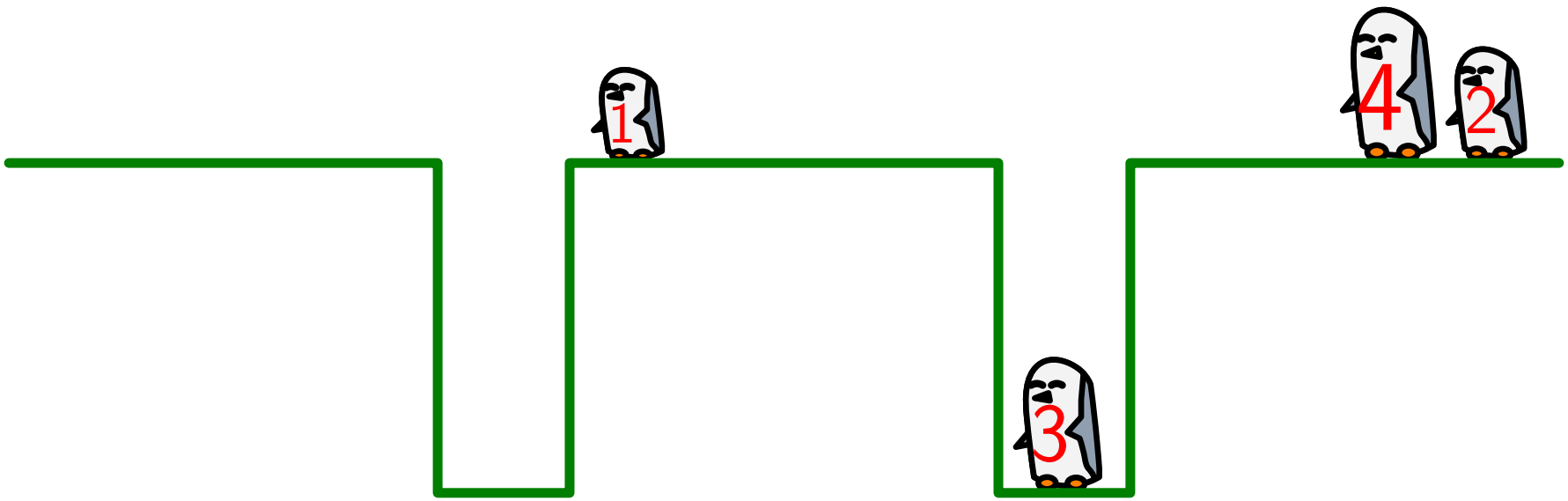
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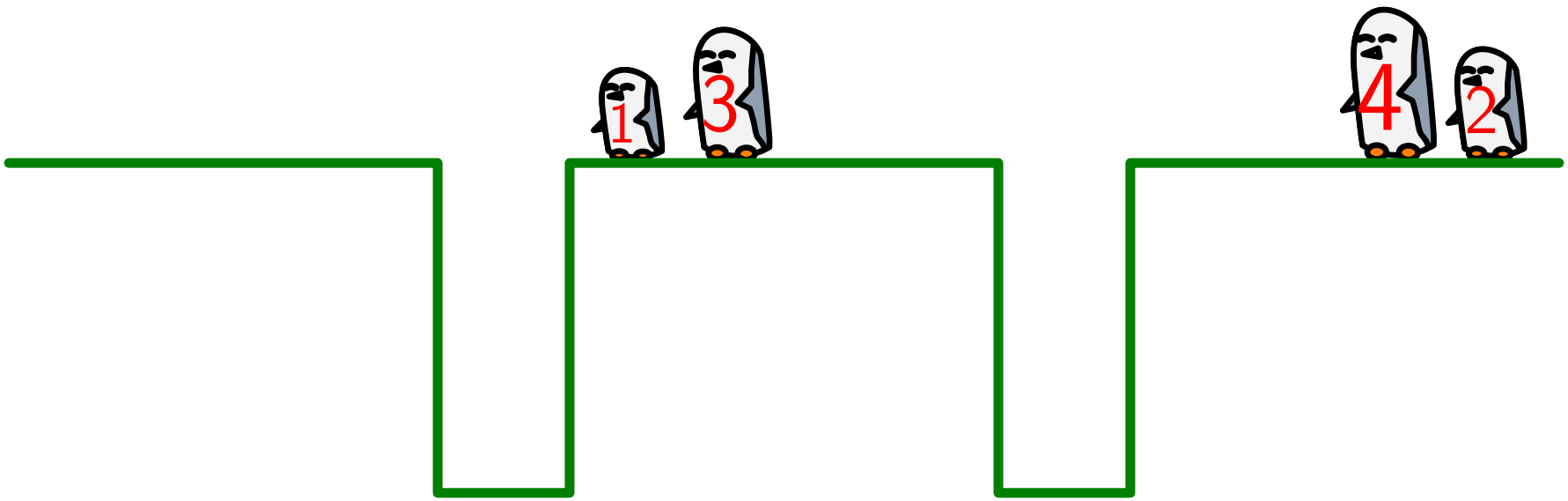
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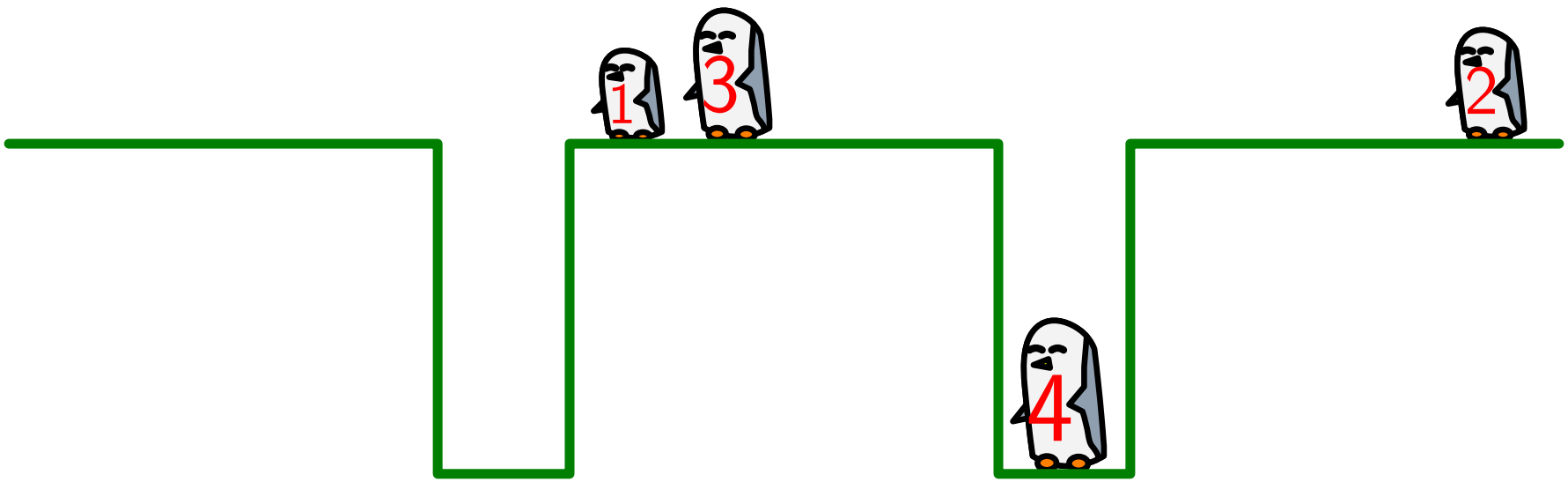
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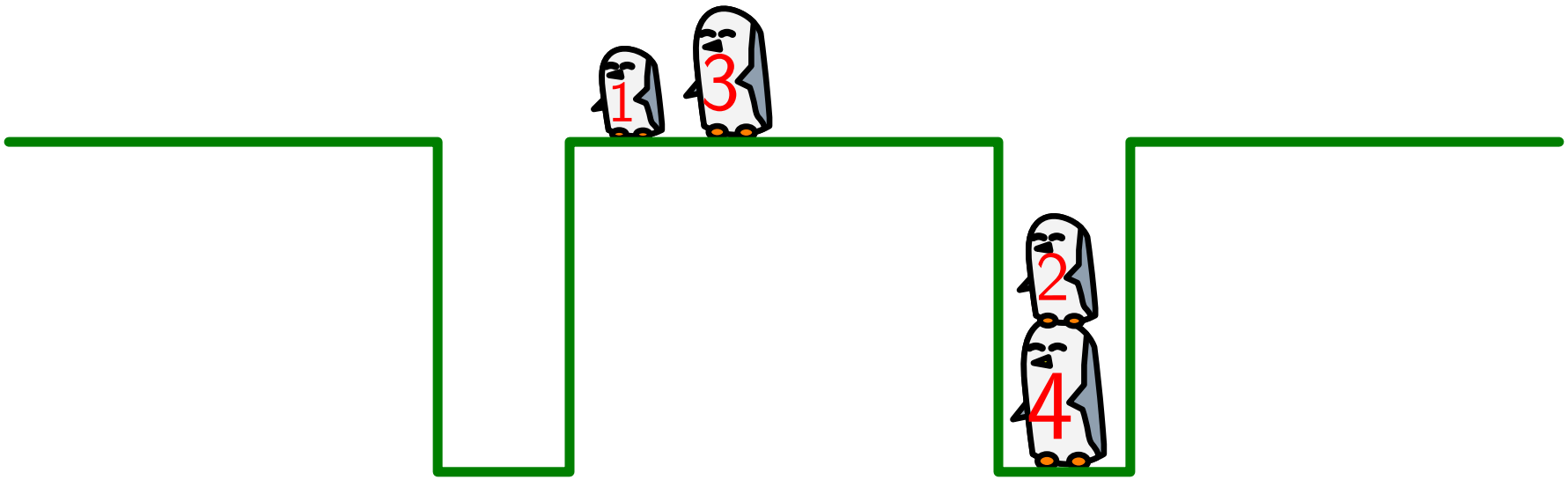
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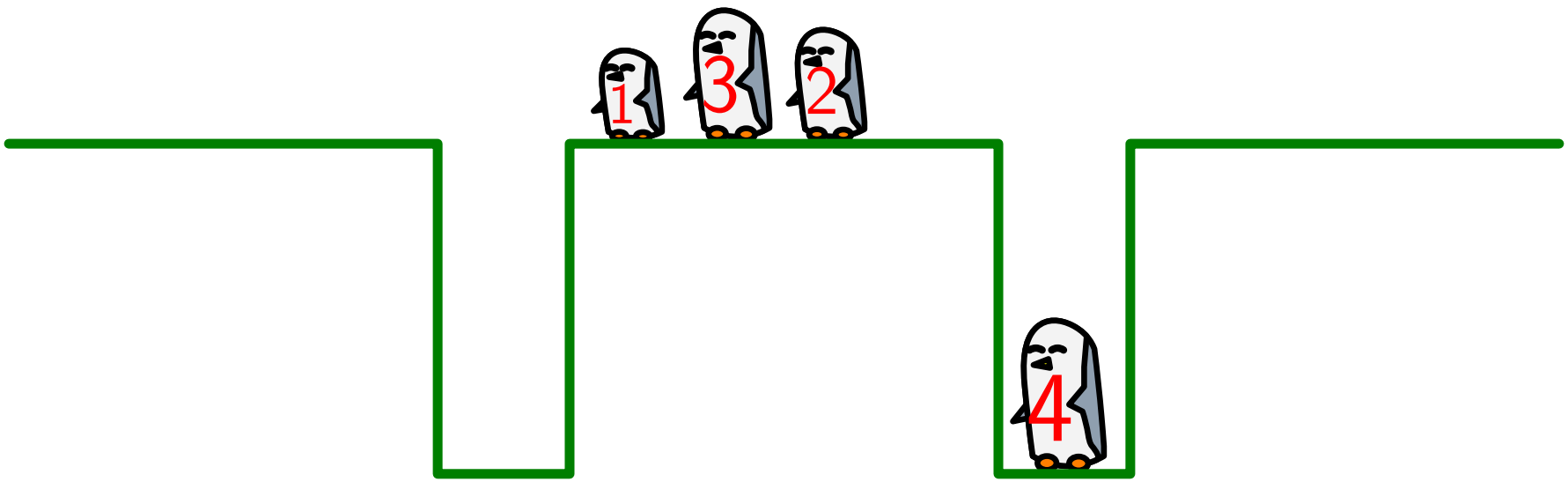
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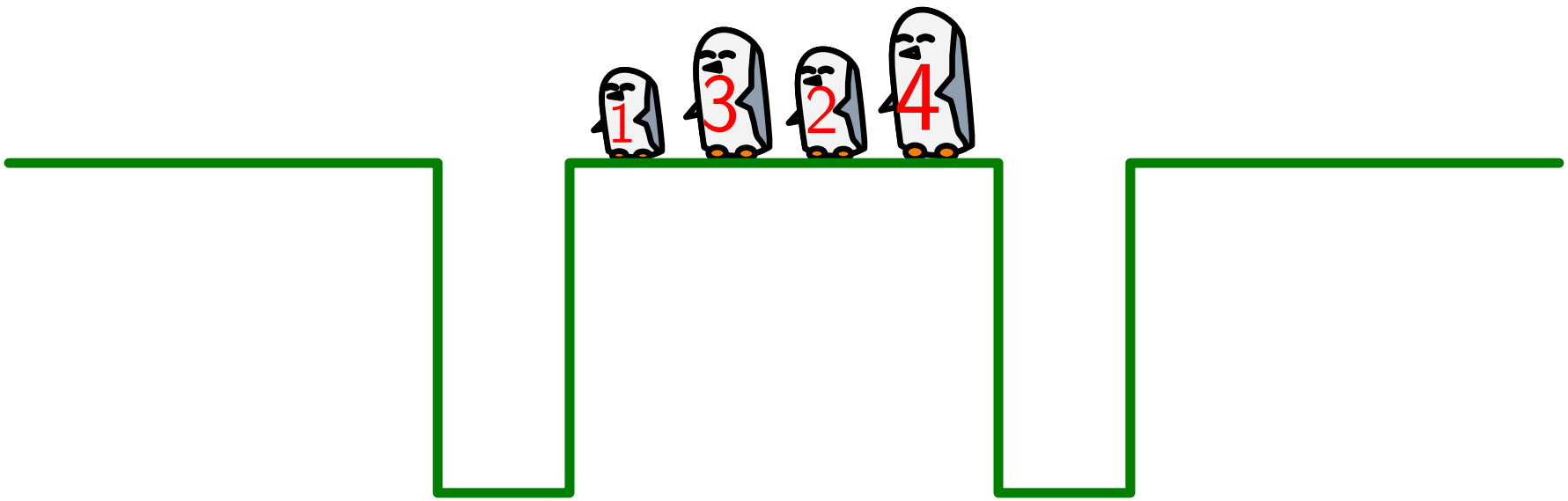
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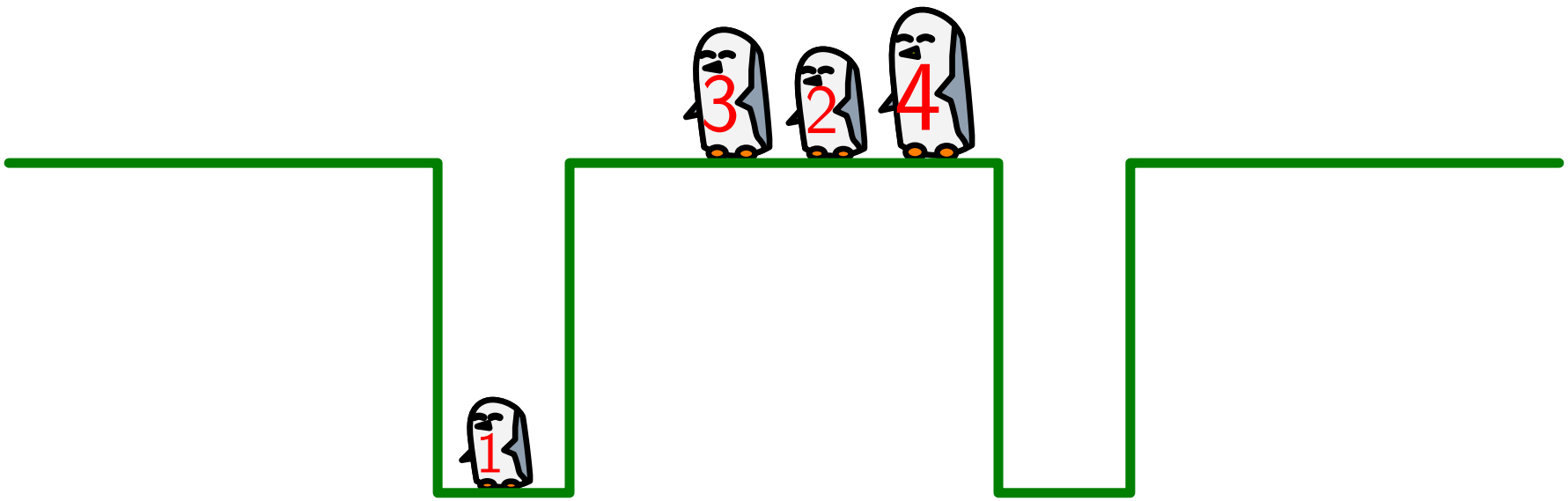
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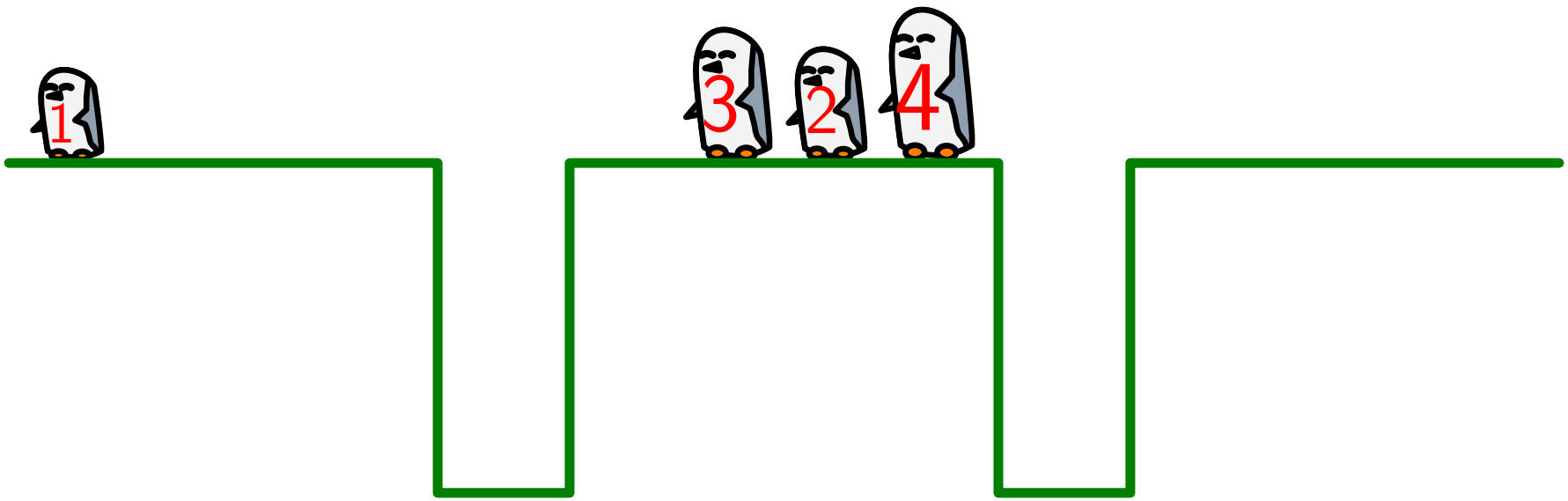
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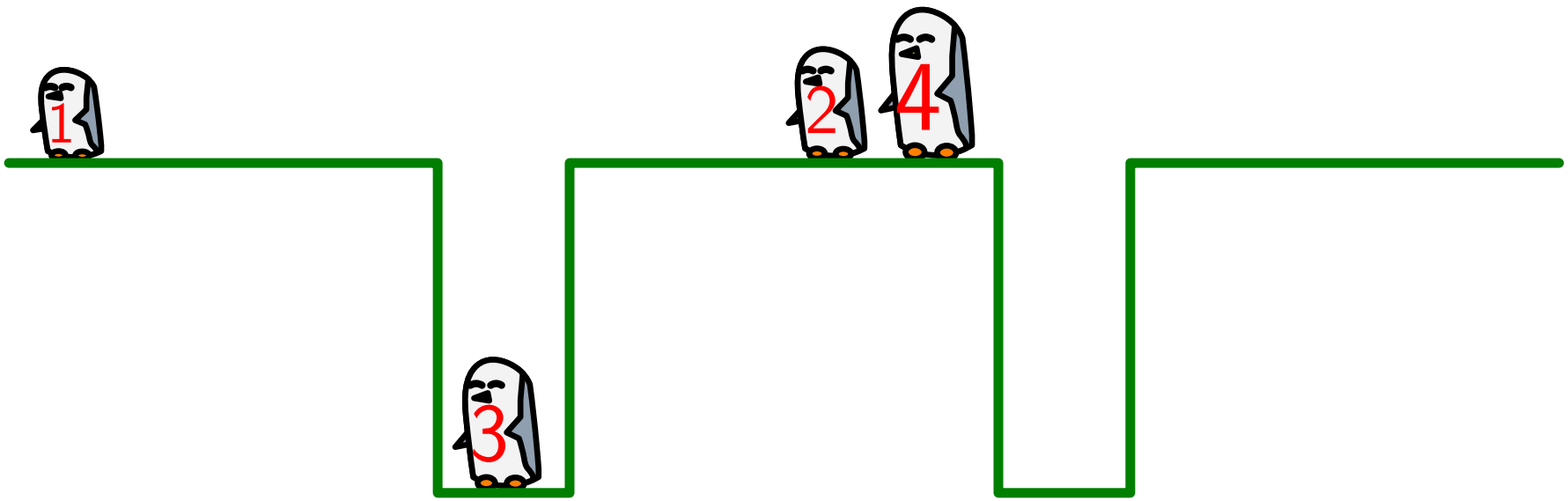
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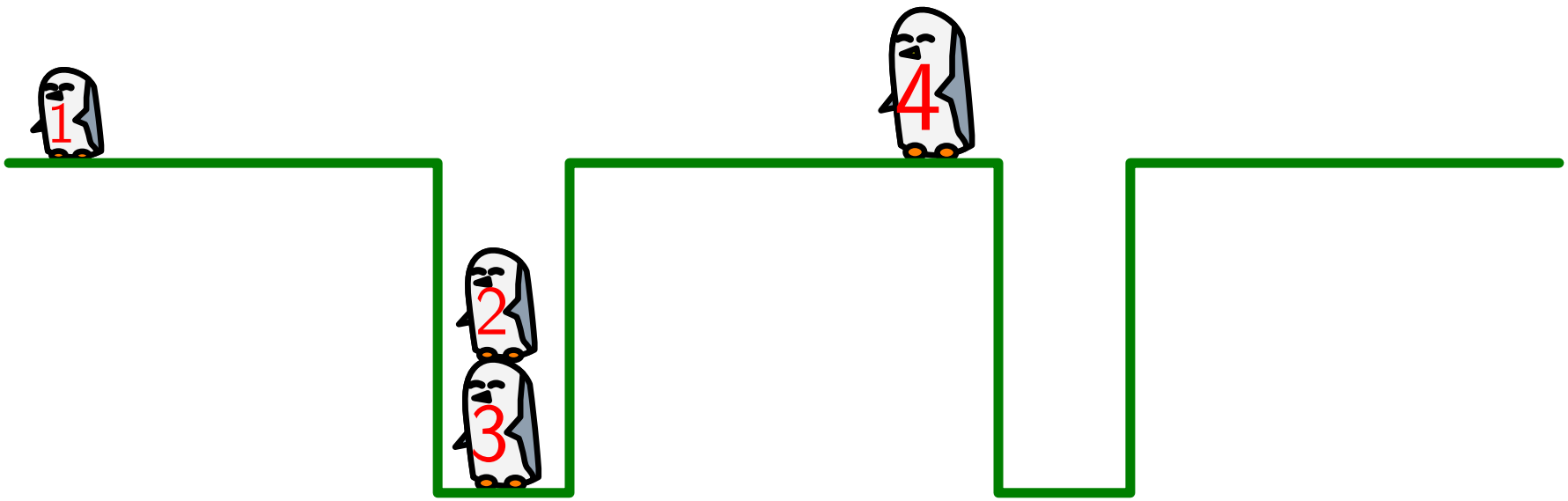
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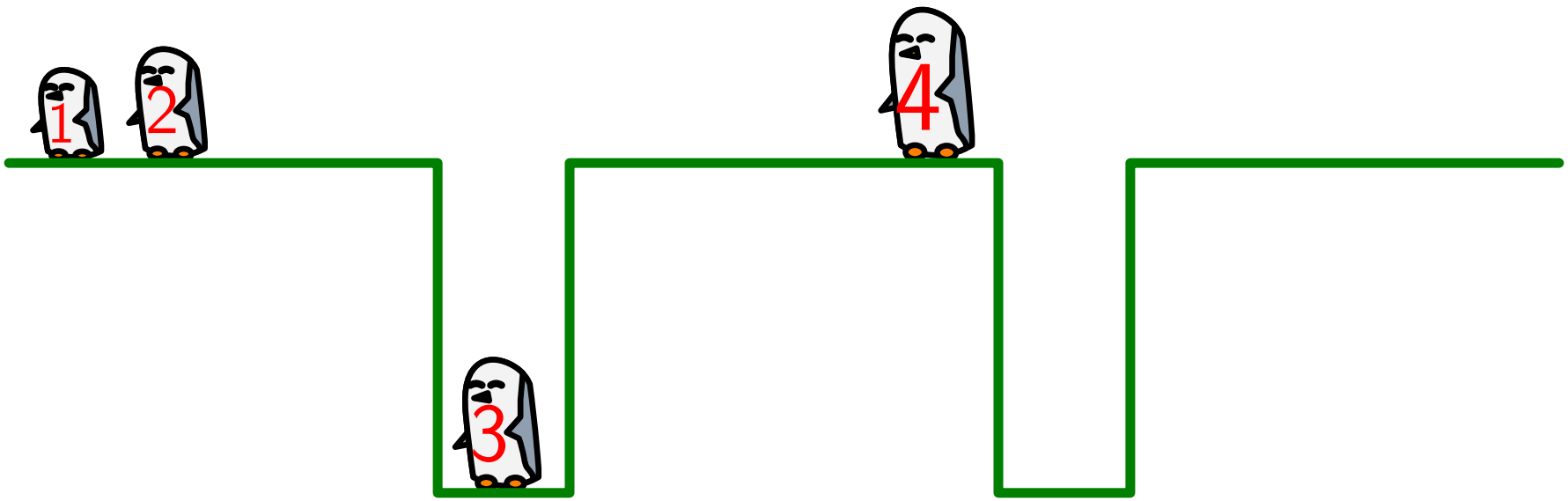
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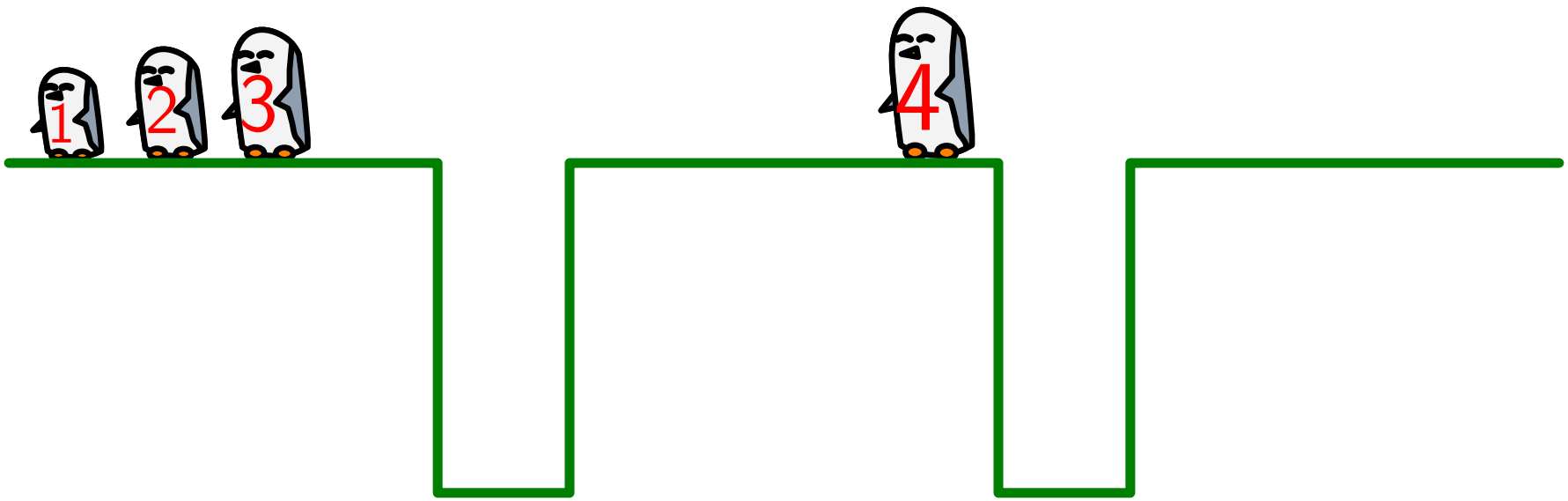
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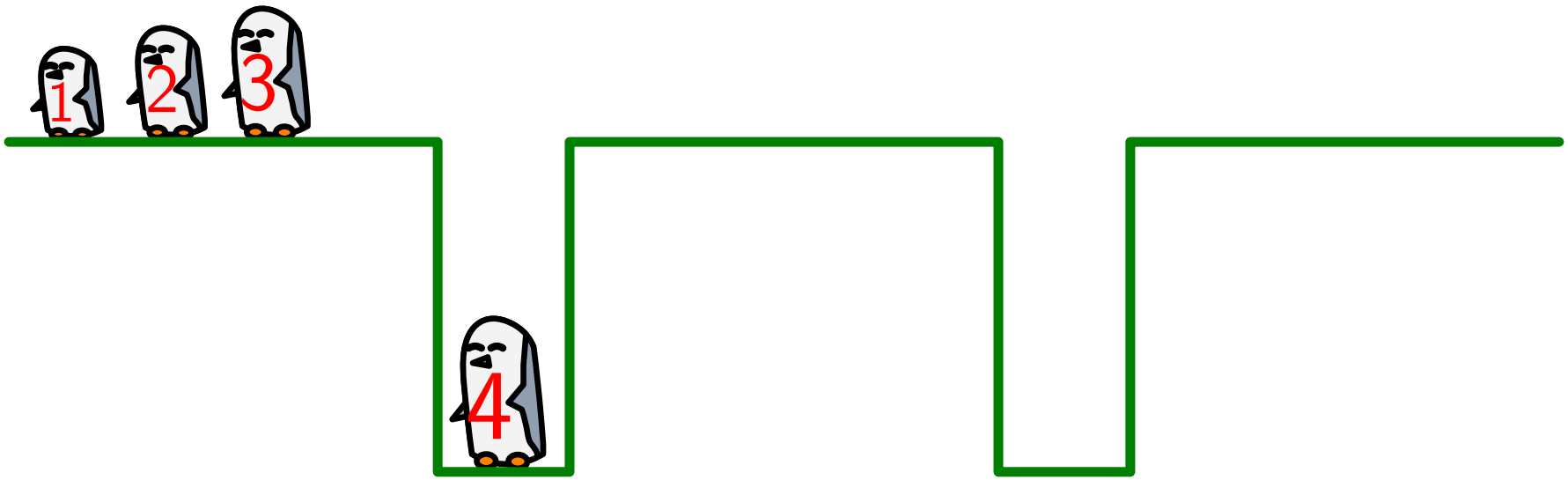
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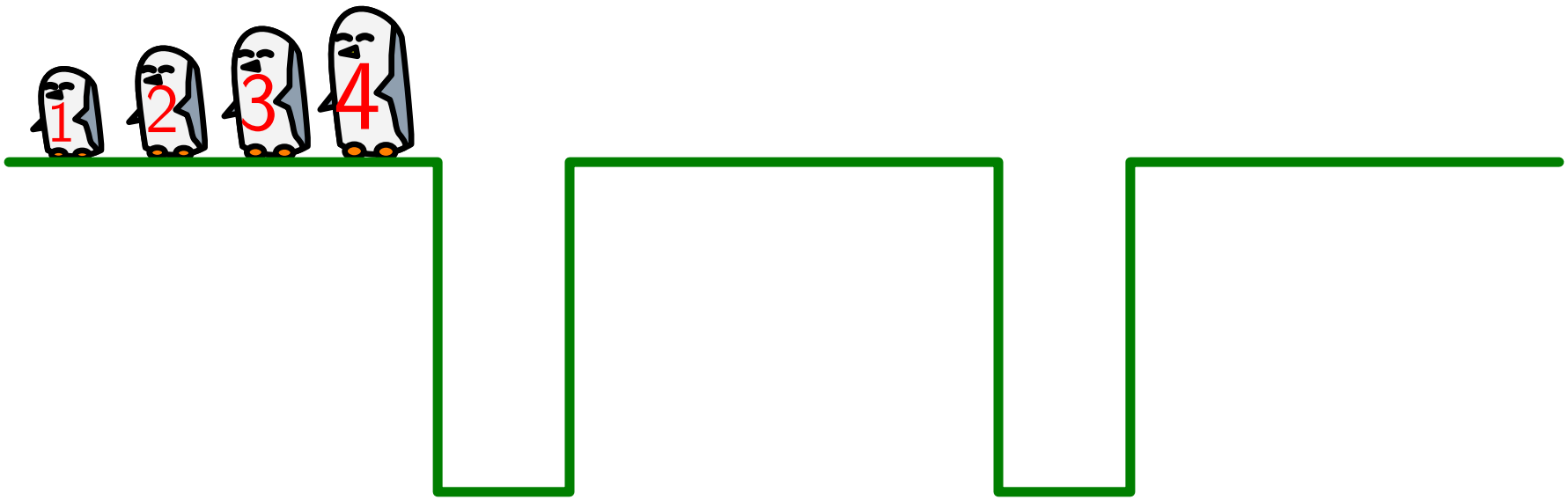
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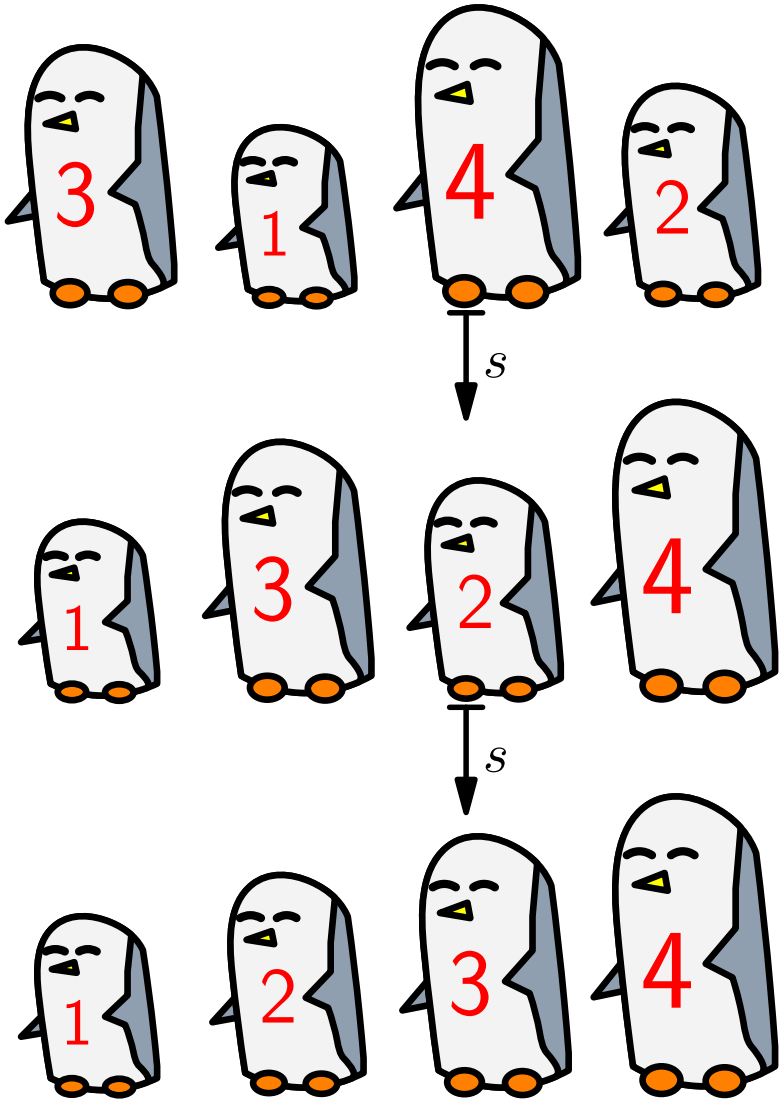
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In notation: $s(3142) = 1324$, $s(1324) = 1234$

Figure generated with AI assistance.

Main result

$$S_3 = \{123, 132, 213, 231, 312, 321\}$$

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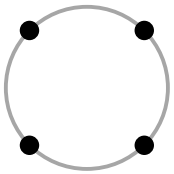
Theorem (L. 2026)

of s^2-sortable permutations in S_n' is nth Motzkin number.

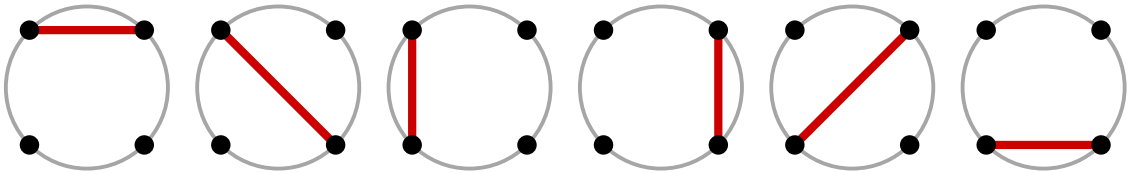
*n*th Motzkin counts:

Ways to draw any # of disjoint chords connecting *n* points on a circle

0 chords:



1 chord:



2 chords:

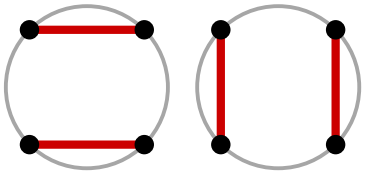
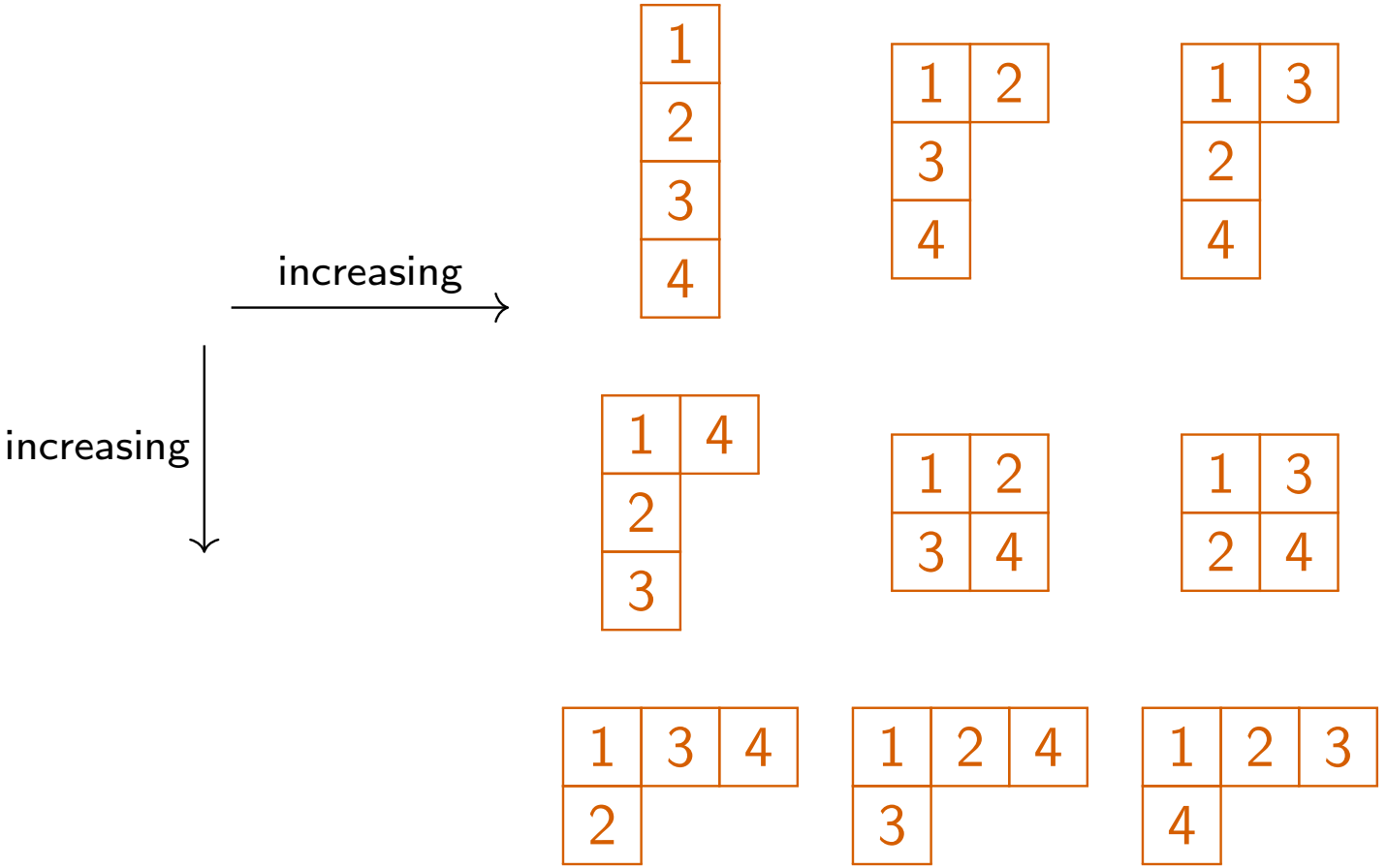


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nth Motzkin counts:

Number of standard Young tableaux with width ≤ 3 (Regev 1981)



*n*th Motzkin counts:

Number of 4321−, (3412,2413)−, (3412,3142)− and 3412−avoiding involutions in S_n.
Number of sequences of length n−1 consisting of positive integers such that the first and last elements are 1 or 2 and the absolute difference between any 2 consecutive elements is 0 or 1. − Jon Perry, Sep 04 2003
From David Callan, Jul 15 2004: (Start)
Also number of Motzkin n−paths: paths from (0,0) to (n,0) in an n X n grid using only steps U = (1,1), F = (1,0) and D = (1,−1).
Number of Dyck n−paths with no UUU. (Given such a Dyck n−path, change each UUD to U, then change each remaining UD to F. This is a bijection to Motzkin n−paths. Example with n=5: U U D U D U U D D D −> U F U D D.)
Number of Dyck (n+1)−paths with no UDU. (Given such a Dyck (n+1)−path, mark each U that is followed by a D and each D that is not followed by a U. Then change each unmarked U whose matching D is marked to an F. Lastly, delete all the marked steps. This is a bijection to Motzkin n−paths. Example with n=6 and marked steps in small type: U U u d D U U u d d d D u d −> U U u d D F F u d d d D u d −> U U D F F D.) (End)
a(n) is the number of strings of length 2n+2 from the following recursively defined set: L contains the empty string and, for any strings a and b in L, we also find (ab) in L. The first few elements of L are e, (), (()), ((())), (()()), (((()))), ((()())), (()(())), (()()()) and so on. This proves that a(n) is less than or equal to C(n), the n−th Catalan number. See Orrick link (2024). − Saul Schleimer (saulsch(AT)math.rutgers.edu), Feb 23 2006 (Additional linked comment added by William P. Orrick, Jun 13 2024.)
a(n) = number of Dyck n−paths all of whose valleys have even x−coordinate (when path starts at origin). For example, T(4,2)=3 counts UDUDUUDD, UDUUDDUD, UUDDUDUD. Given such a path, split it into n subpaths of length 2 and transform UU−>U, DD−>D, UD−>F (there will be no DUs for that would entail a valley with odd x−coordinate). This is a bijection to Motzkin n−paths. − David Callan, Jun 07 2006
Also the number of standard Young tableaux of height <= 3. − Mike Zabrocki, Mar 24 2007
a(n) is the number of RNA shapes of size 2n+2. RNA Shapes are essentially Dyck words without "directly nested" motifs of the form A[[B]]C, for A, B and C Dyck words. The first RNA Shapes are []; [][]; [][][], [[]][]; [][][][], [][[]][], [[][]][], [[]][][]; ... − Yann Ponty (ponty(AT)lri.fr), May 30 2007
The sequence is self−generated from top row A going to the left starting (1,1) and bottom row = B, the same sequence but starting (0,1) and going to the right. Take dot product of A and B and add the result to n−th term of A to get the (n+1)−th term of A. Example: a(5) = 21 as follows: Take dot product of A = (9, 4, 2, 1, 1) and (0, 1, 1, 2, 4) = (0, + 4 + 2 + 2 + 4) = 12; which is added to 9 = 21. − Gary W. Adamson, Oct 27 2008
Equals A005773 / A005773 shifted (i.e., (1,2,5,13,35,96,...) / (1,1,2,5,13,35,96,...)). − Gary W. Adamson, Dec 21 2008
Starting with offset 1 = iterates of M * [1,1,0,0,0,...], where M = a tridiagonal matrix with [0,1,1,1,...] in the main diagonal and [1,1,1,...] in the super and subdiagonals. − Gary W. Adamson, Jan 07 2009
a(n) is the number of involutions of {1,2,...,n} having genus 0. The genus g(p) of a permutation p of {1,2,...,n} is defined by g(p)=(1/2)[n+1−z(p)−z(cp')], where p' is the inverse permutation of p, c = 234...n1 = (1,2,...,n), and z(q) is the number of cycles of the permutation q. Example: a(4)=9; indeed, p=3412=(13)(24) is the only involution of {1,2,3,4} with genus > 0. This follows easily from the fact that a permutation p of {1,2,...,n} has genus 0 if and only if the cycle decomposition of p gives a noncrossing partition of {1,2,...,n} and each cycle of p is increasing (see Lemma 2.1 of the Dulucq−Simion reference). [Also, redundantly, for p=3412=(13)(24) we have cp'=2341*3412=4123=(1432) and so g(p)=(1/2)(4+1−2−1)=1.] − Emeric Deutsch, May 29 2010
Let w(i,j,n) denote walks in N^2 which satisfy the multivariate recurrence w(i,j,n) = w(i, j + 1, n − 1) + w(i − 1, j, n − 1) + w(i + 1, j − 1, n − 1) with boundary conditions w(0,0,0) = 1 and w(i,j,n) = 0 if i or j or n is < 0. Then a(n) = Sum_{i = 0..n, j = 0..n} w(i,j,n) is the number of such walks of length n. − Peter Luschny, May 21 2011
a(n)/a(n−1) tends to 3.0 as N−>infinity: (1+2*cos(2*Pi/N)) relating to longest odd N regular polygon diagonals, by way of example, N=7: Using the tridiagonal generator [cf. comment of Jan 07 2009], for polygon N=7, we extract an (N−1)/2 = 3 X 3 matrix, [0,1,0; 1,1,1; 0,1,1] with an e−val of 2.24697...; the longest Heptagon diagonal with edge = 1. As N tends to infinity, the diagonal lengths tend to 3.0, the convergent of the sequence. − Gary W. Adamson, Jun 08 2011
Number of (n+1)−length permutations avoiding the pattern 132 and the dotted pattern 23\dot{1}. − Jean−Luc Baril, Mar 07 2012
Number of n−length words w over alphabet {a,b,c} such that for every prefix z of w we have #(z,a) >= #(z,b) >= #(z,c), where #(z,x) counts the letters x in word z. The a(4) = 9 words are: aaaa, aaab, aaba, abaa, aabb, abab, aabc, abac, abca. − Alois P. Heinz, May 26 2012
Number of length−n restricted growth strings (RGS) [r(1), r(2), ..., r(n)] such that r(1)=1, r(k)<=k, and r(k)!=r(k−1); for example, the 9 RGS for n=4 are 1010, 1012, 1201, 1210, 1212, 1230, 1231, 1232, 1234. − Joerg Arndt, Apr 16 2013
Number of length−n restricted growth strings (RGS) [r(1), r(2), ..., r(n)] such that r(1)=0, r(k)<=k and r(k)−r(k−1) != 1; for example, the 9 RGS for n=4 are 0000, 0002, 0003, 0004, 0022, 0024, 0033, 0222, 0224. − Joerg Arndt, Apr 17 2013
Number of (4231,5276143)−avoiding involutions in S_n. − Alexander Burstein, Mar 05 2014
a(n) is the number of increasing unary−binary trees with n nodes that have an associated permutation that avoids 132. For more information about unary−binary trees with associated permutations, see A245888. − Manda Riehl, Aug 07 2014

From Tony Foster III, Jul 28 2016: (Start)
A series created using 2*a(n) + a(n+1) has Hankel transform of F(2n), offset 3, F being the Fibonacci bisection, A001906 (empirical observation).
A series created using 2*a(n) + 3*a(n+1) + a(n+2) gives the Hankel transform of Sum_{k=0..n} k*Fibonacci(2*k), offset 3, A197649 (empirical observation). (End)
Conjecture: (2/n)*Sum_{k=1..n} (2k+1)*a(k)^2 is an integer for each positive integer n. − Zhi−Wei Sun, Nov 16 2017
The Rubey and Stump reference proves a refinement of a conjecture of René Marczinzik, which they state as: "The number of 2−Gorenstein algebras which are Nakayama algebras with n simple modules and have an oriented line as associated quiver equals the number of Motzkin paths of length n." − Eric M. Schmidt, Dec 16 2017
Number of U_{k}−equivalence classes of Łukasiewicz paths. Łukasiewicz paths are P−equivalent iff the positions of pattern P are identical in these paths. − Sergey Kirgizov, Apr 08 2018
If tau_1 and tau_2 are two distinct permutation patterns chosen from the set {132,231,312}, then a(n) is the number of valid hook configurations of permutations of [n+1] that avoid the patterns tau_1 and tau_2. − Colin Defant, Apr 28 2019
Number of permutations of length n that are sorted to the identity by a consecutive−321−avoiding stack followed by a classical−21−avoiding stack. − Colin Defant, Aug 29 2020
From Helmut Prodinger, Dec 13 2020: (Start)
a(n) is the number of paths in the first quadrant starting at (0,0) and consisting of n steps from the infinite set {(1,1), (1,−1), (1,−2), (1,−3), ...}.
For example, denoting U=(1,1), D=(1,−1), D_ j=(1,−j) for j >= 2, a(4) counts UUUU, UUUD, UUUD_2, UUUD_3, UUDU, UUDD, UUD_2U, UDUU, UDUD.
This step set is inspired by {(1,1), (1,−1), (1,−3), (1,−5), ...}, suggested by Emeric Deutsch around 2000.
See Prodinger link that contains a bijection to Motzkin paths. (End)
Named by Donaghey (1977) after the Israeli−American mathematician Theodore Motzkin (1908−1970). In Sloane's "A Handbook of Integer Sequences" (1973) they were called "generalized ballot numbers". − Amiram Eldar, Apr 15 2021
Number of Motzkin n−paths a(n) is split into A107587(n), number of even Motzkin n−paths, and A343386(n), number of odd Motzkin n−paths. The value A107587(n) − A343386(n) can be called the "shadow" of a(n) (see A343773). − Gennady Eremin, May 17 2021
Conjecture: If p is a prime of the form 6m+1 (A002476), then a(p−2) is divisible by p. Currently, no counterexample exists for p < 10^7. Personal communication from Robert Gerbicz: mod such p this is equivalent to A066796 with comment: "Every A066796(n) from A066796((p−1)/2) to A066796(p−1) is divisible by prime p of form 6m+1". − Serge Batalov, Feb 08 2022
From Rob Burns, Nov 11 2024: (Start)
The conjecture is proved in the 2017 paper by Rob Burns in the Links below. The result is contained in Tables 4 and 5 of the paper, which show that a(p−2) == 0 (mod p) when p == 1 (mod 6) and a(p−2) == −1 (mod p) when p == −1 (mod 6).
In fact, the 2017 paper by Burns establishes more general congruences for a(p^k − 2) where k >= 1.
If p == 1 (mod 6) then a(p^k − 2) == 0 (mod p) for k >= 1.
If p == −1 (mod 6) then a(p^k − 2) == −1 (mod p) when k is odd and a(p^k − 2) == 0 (mod p) when k is even.
These are consequences of the transitions provided in Tables 4, 5 and 6 of the paper.
The 2024 paper by Nadav Kohen also proves the conjecture. Proposition 6 of the paper states that a prime p divides a(p−2) if and only if p = (1 mod 3). (End)
From Peter Bala, Feb 10 2022: (Start)
Conjectures:
(1) For prime p == 1 (mod 6) and n, r >= 1, a(n*p^r − 2) == −A005717(n−1) (mod p), where we take A005717(0) = 0 to match Batalov's conjecture above.
(2) For prime p == 5 (mod 6) and n >= 1, a(n*p − 2) == −A005773(n) (mod p).
(3) For prime p >= 3 and k >= 1, a(n + p^k) == a(n) (mod p) for 0 <= n <= (p^k − 3).
(4) For prime p >= 5 and k >= 2, a(n + p^k) == a(n) (mod p^2) for 0 <= n <= (p^(k−1) − 3). (End)
The Hankel transform of this sequence with a(0) omitted gives the period−6 sequence [1, 0, −1, −1, 0, 1, ...] which is A010892 with its first term omitted, while the Hankel transform of the current sequence is the all−ones sequence A000012, and also it is the unique sequence with this property which is similar to the unique Hankel transform property of the Catalan numbers. − Michael Somos, Apr 17 2022
The number of terms in which the exponent of any variable x_i is not greater than 2 in the expansion of Product_{j=1..n} Sum_{i=1..j} x_i. E.g.: a(4) = 9: 3*x1^2*x2^2, 4*x1^2*x2*x3, 2*x1^2*x2*x4, x1^2*x3^2, x1^2*x3*x4, 2*x1*x2^2*x3, x1*x2^2*x4, x1*x2*x3^2, x1*x2*x3*x4. − Elif Baser, Dec 20 2024
The Hankel matrix corresponding to this sequence factorizes as T * Transpose(T), where T is the lower unitriangular array A064189 (row reverse of the Motzkin triangle A026300). Hence the Hankel determinants are all equal to 1 as stated above by Michael Somos. − Peter Bala, Nov 23 2025
From Michael Luo, Jul 29 2026: (Start)
Number of 2−stack−sortable permutations of {1,2,...,n + 1} ending with 1.
Number of standard composition tableaux of width at most 2 on n cells. (End)

Proof idea

Theorem (L. 2026)

of s^2-sortable permutations in S'_n is nth Motzkin number.

Idea: matching!

Figure generated with AI assistance.

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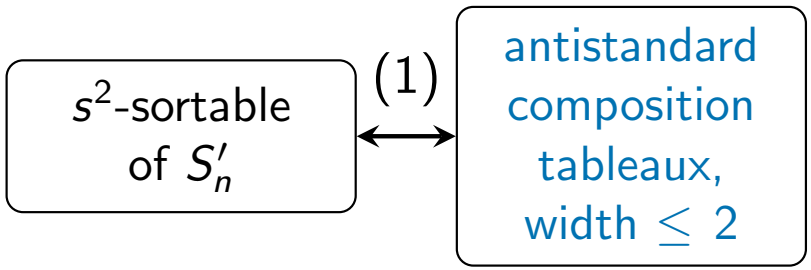


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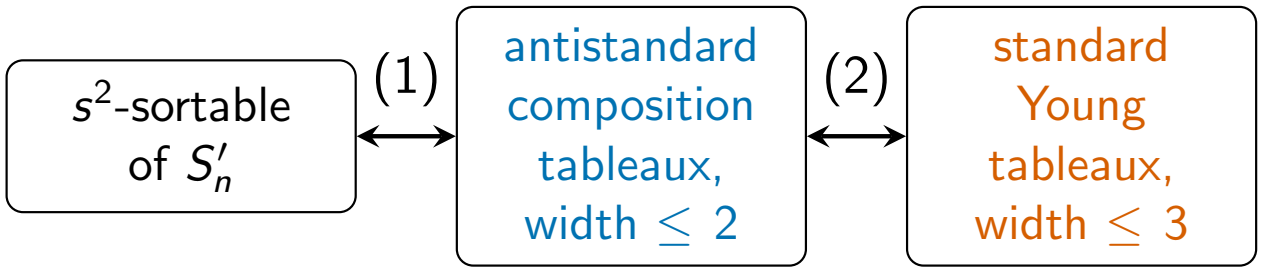


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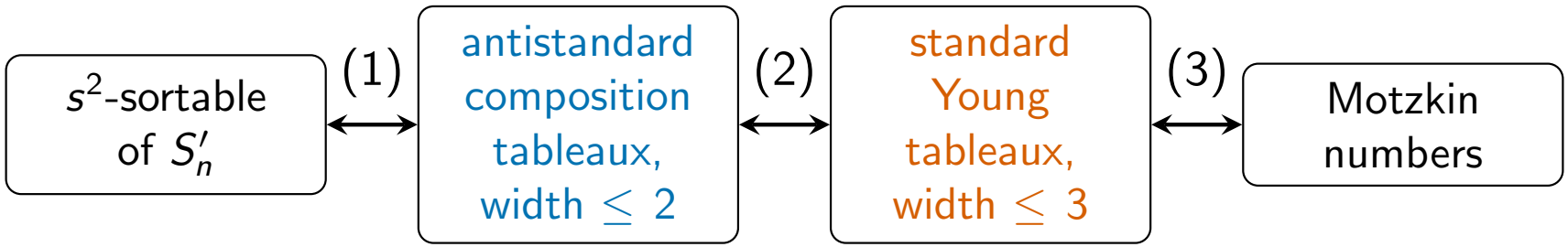


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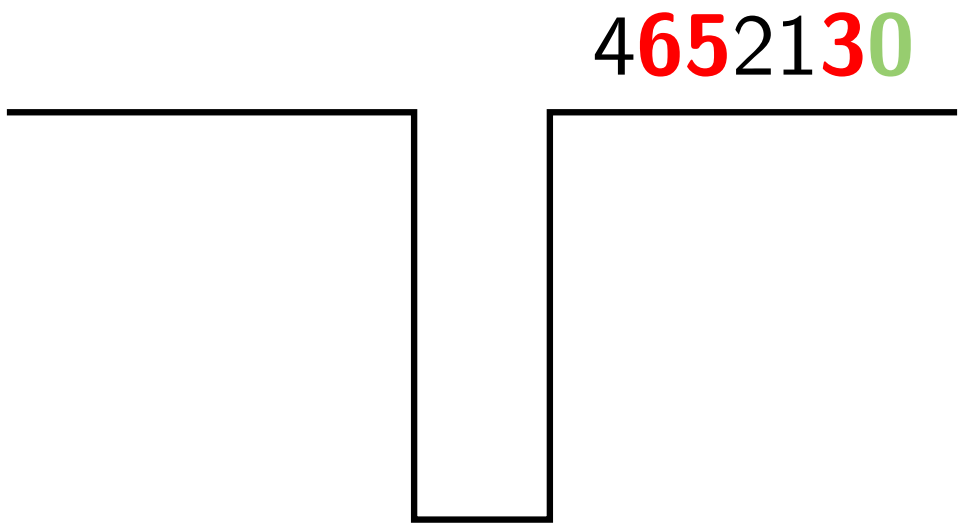
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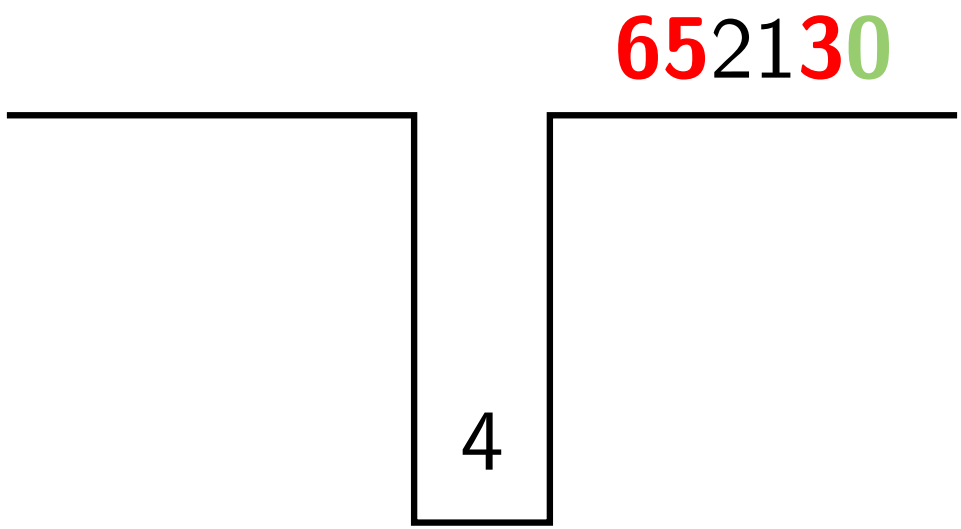


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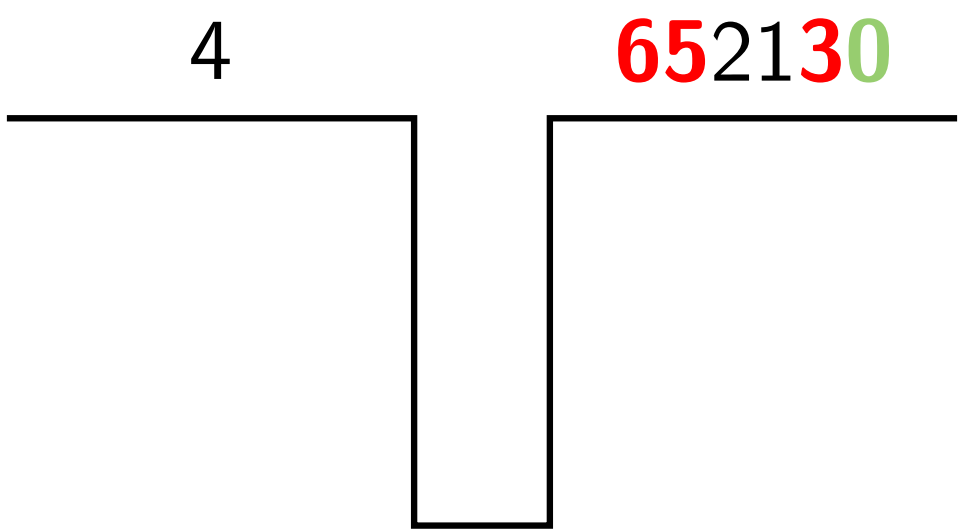


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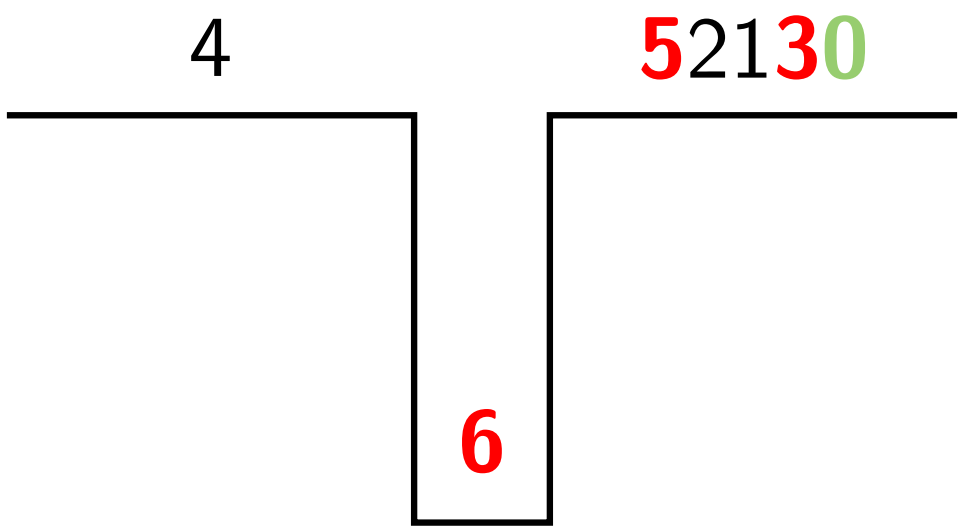


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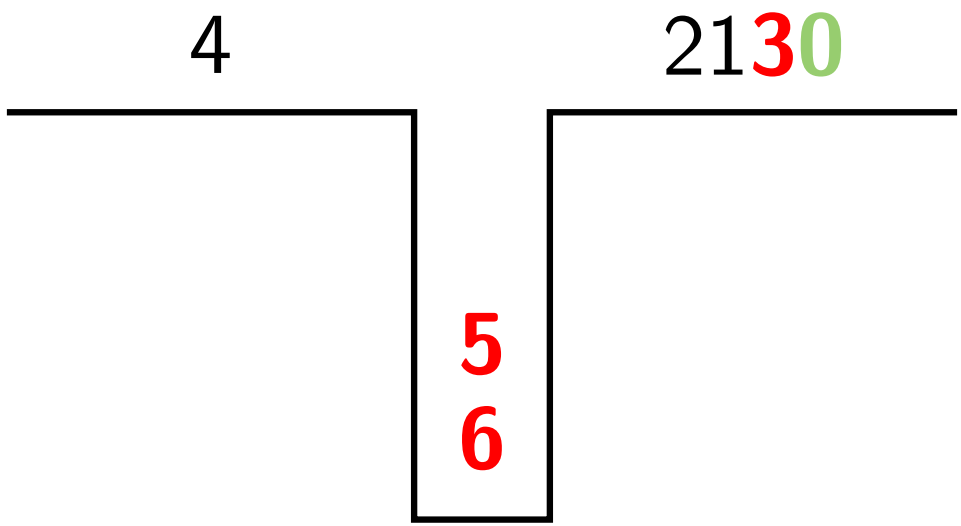


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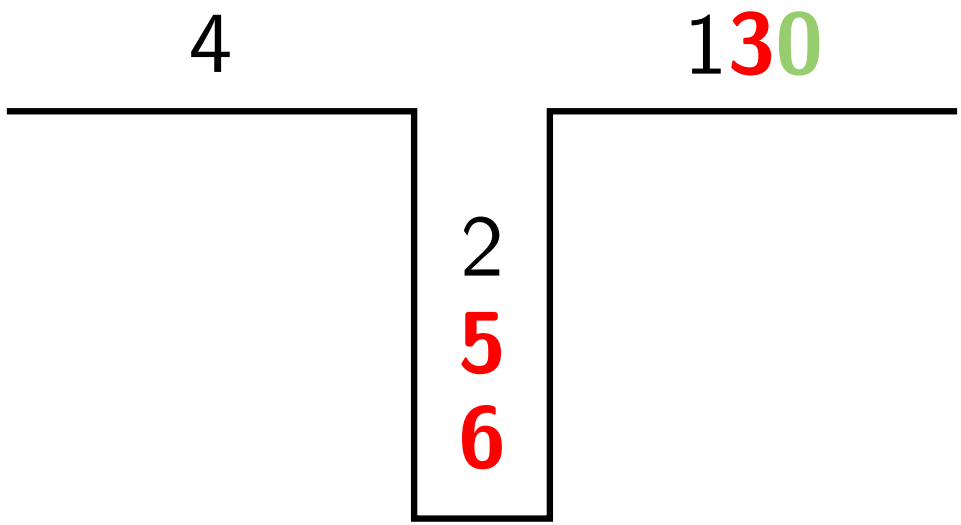


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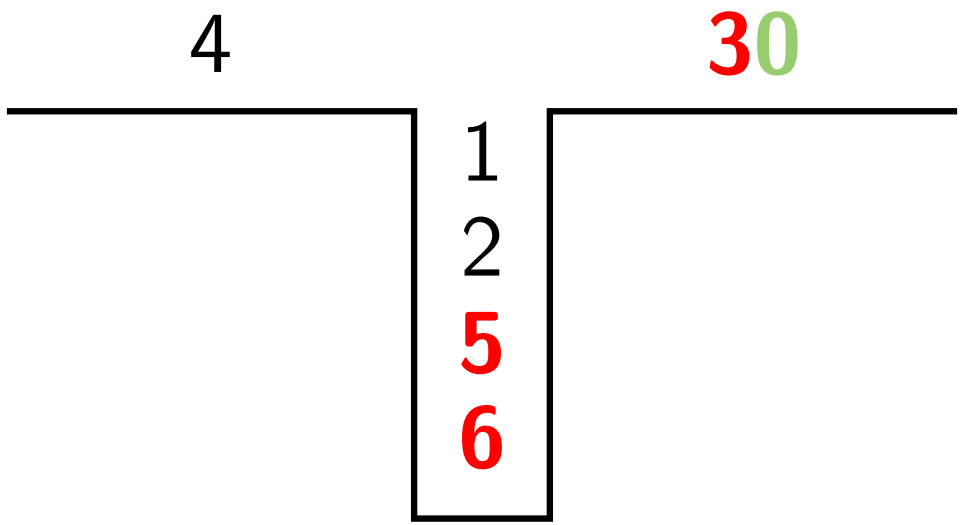


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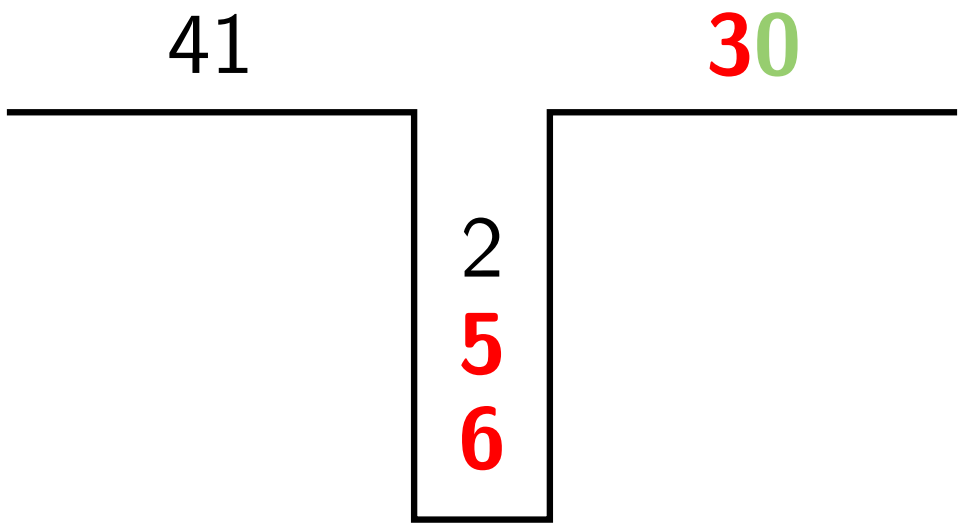


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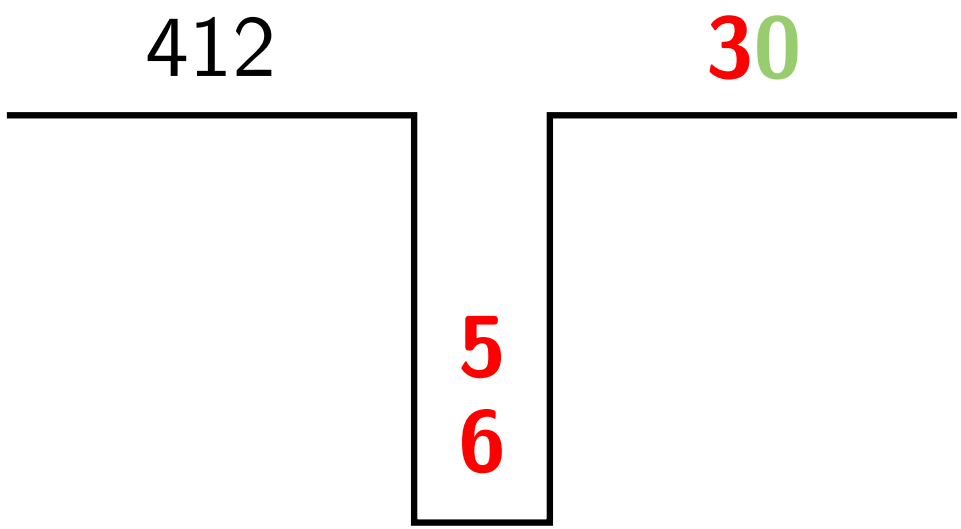


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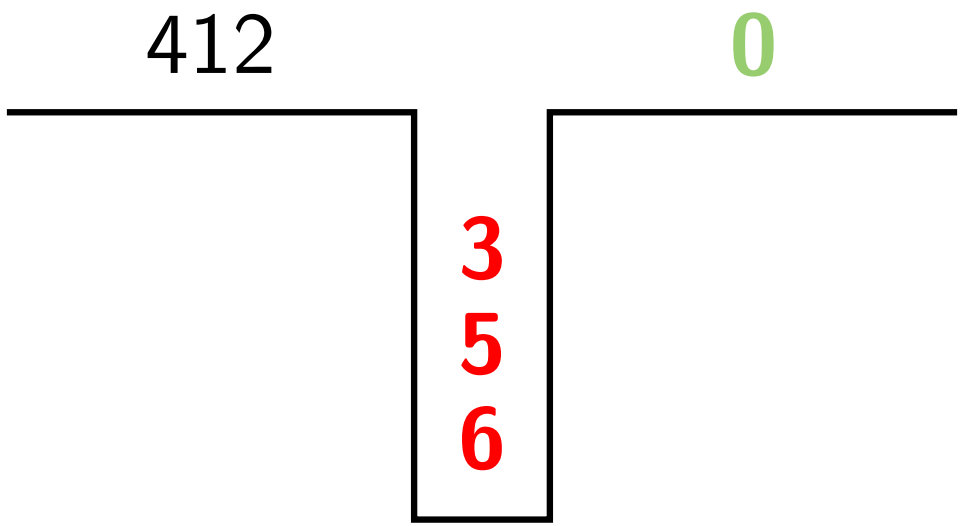


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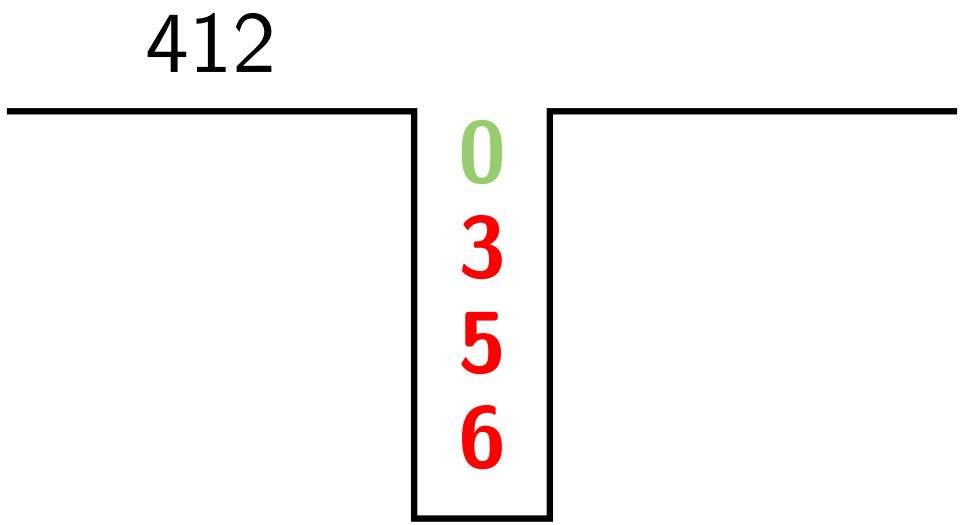


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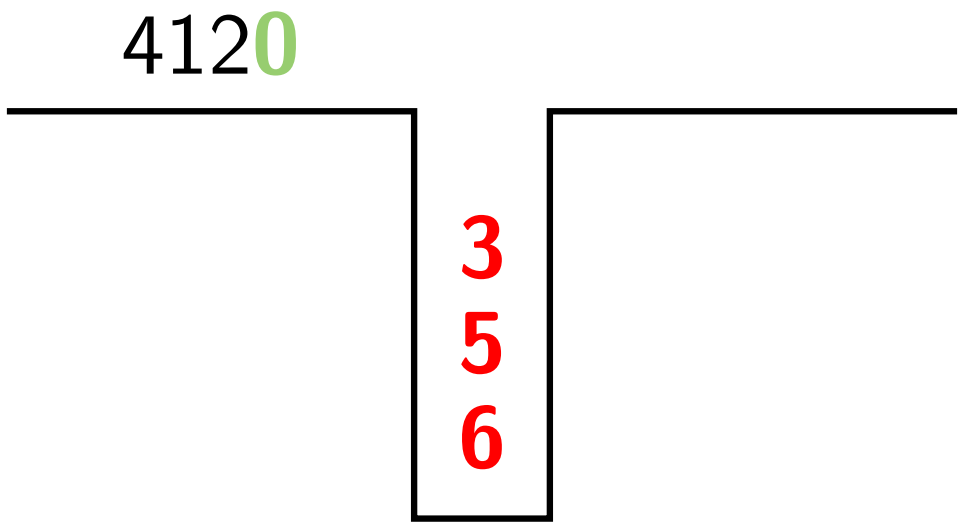


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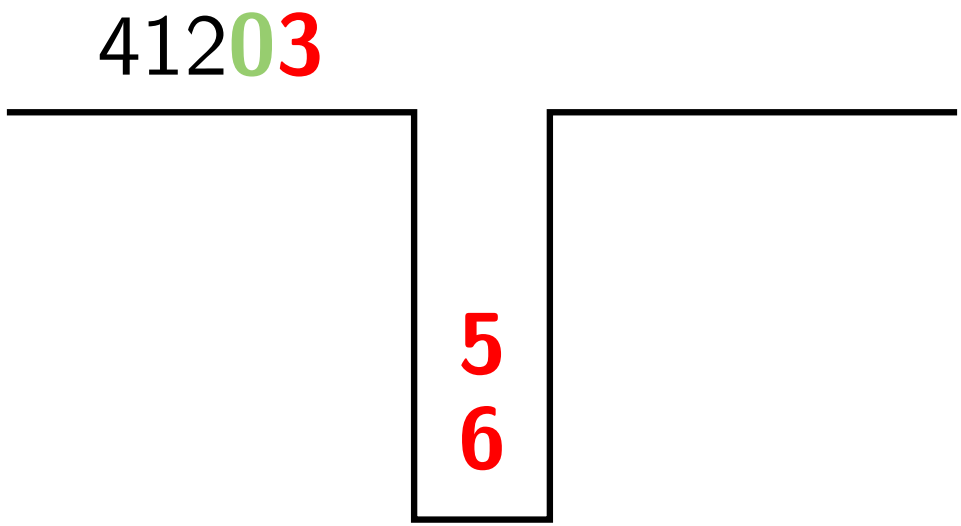


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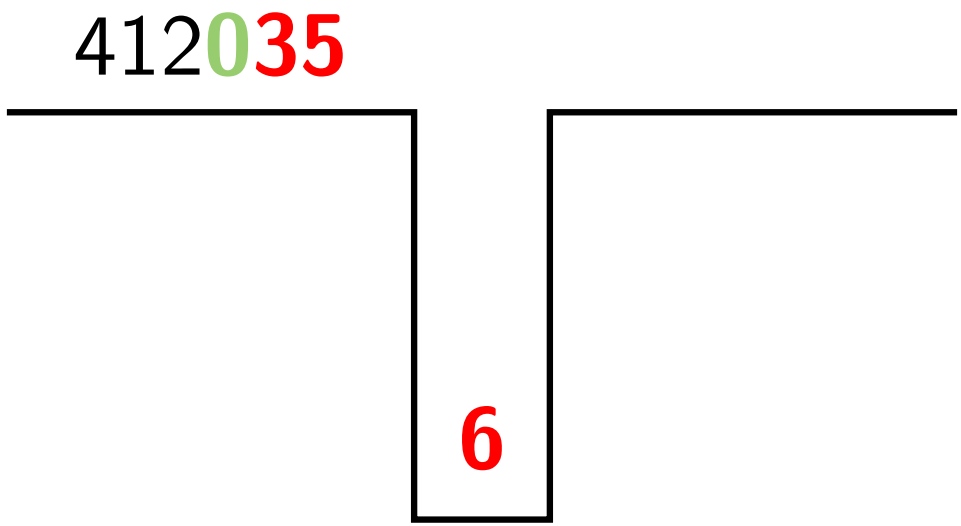


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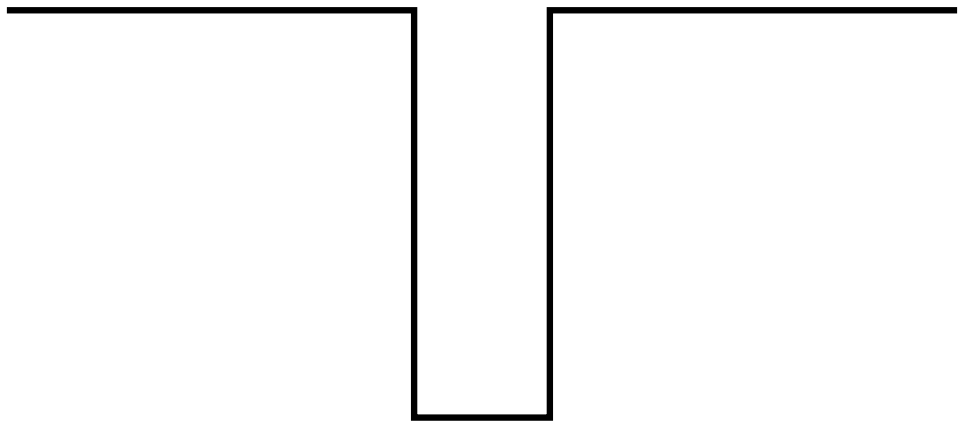
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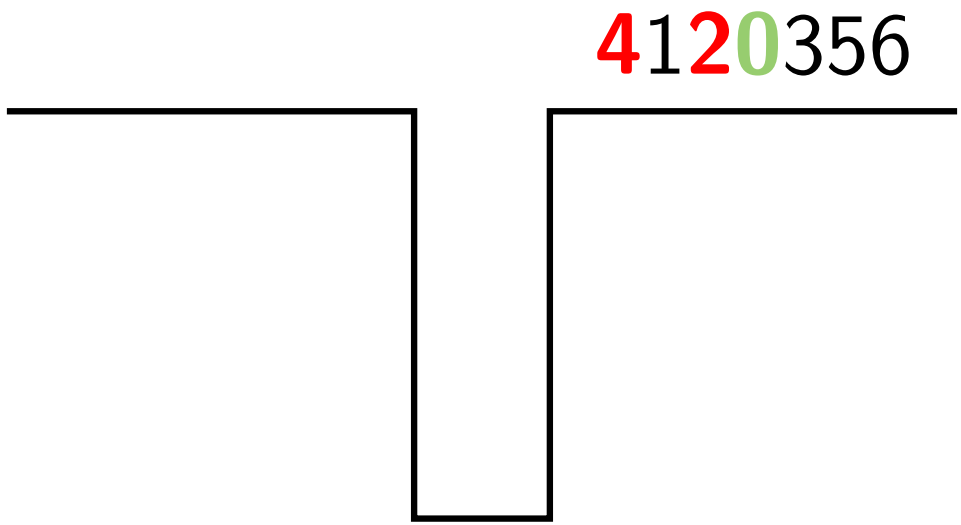


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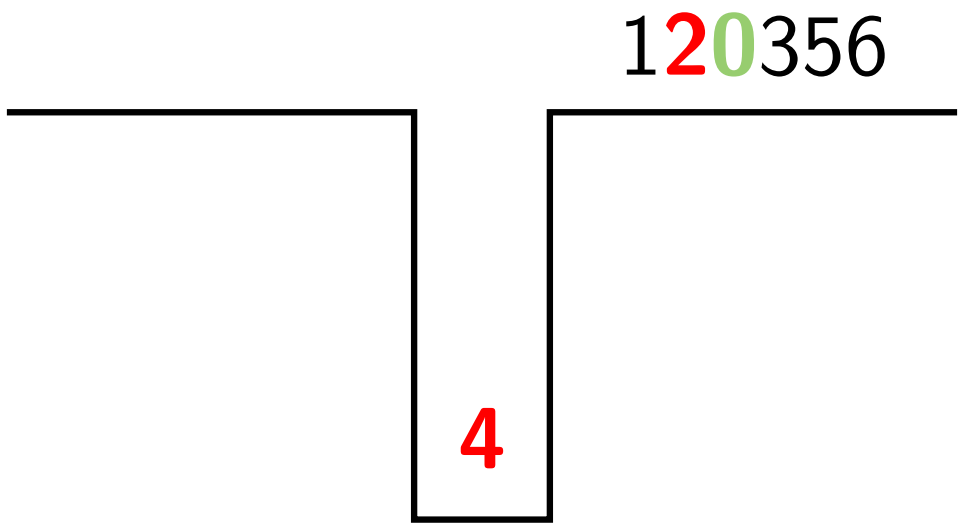


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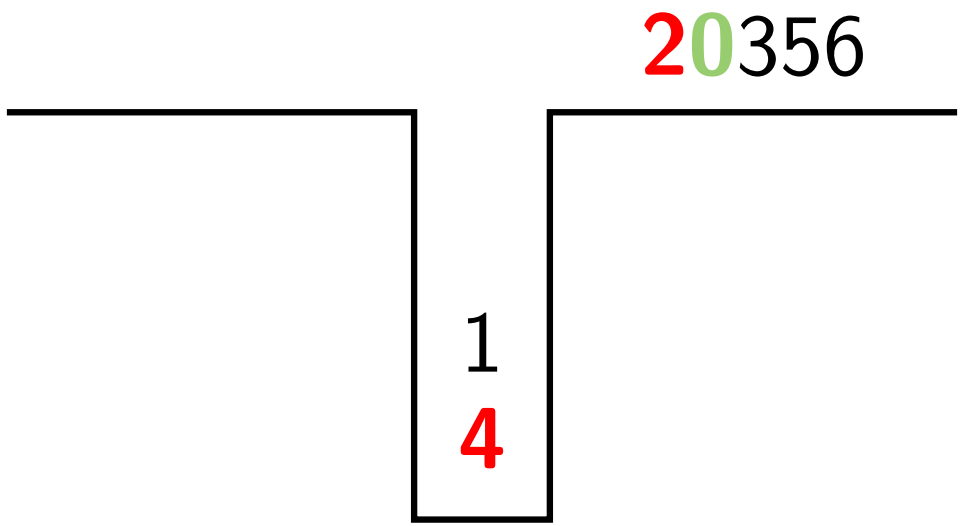


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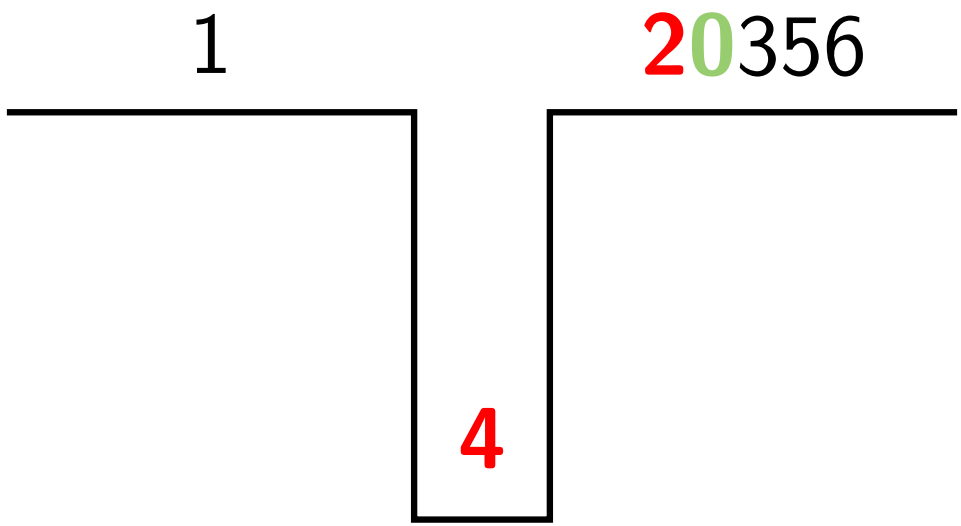


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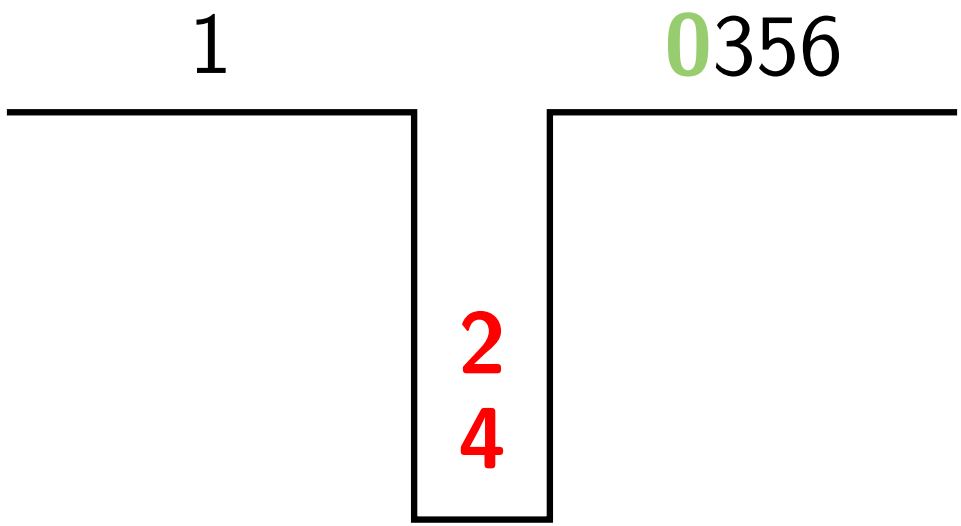


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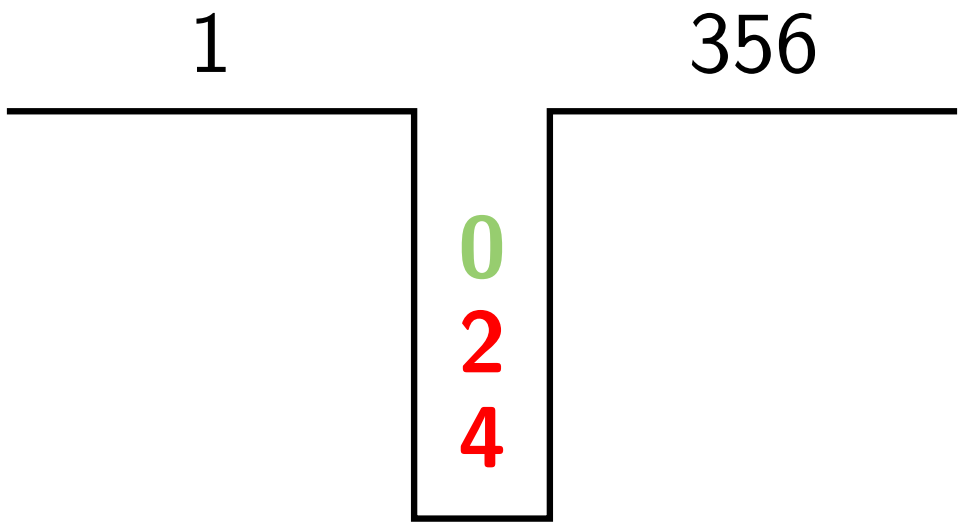


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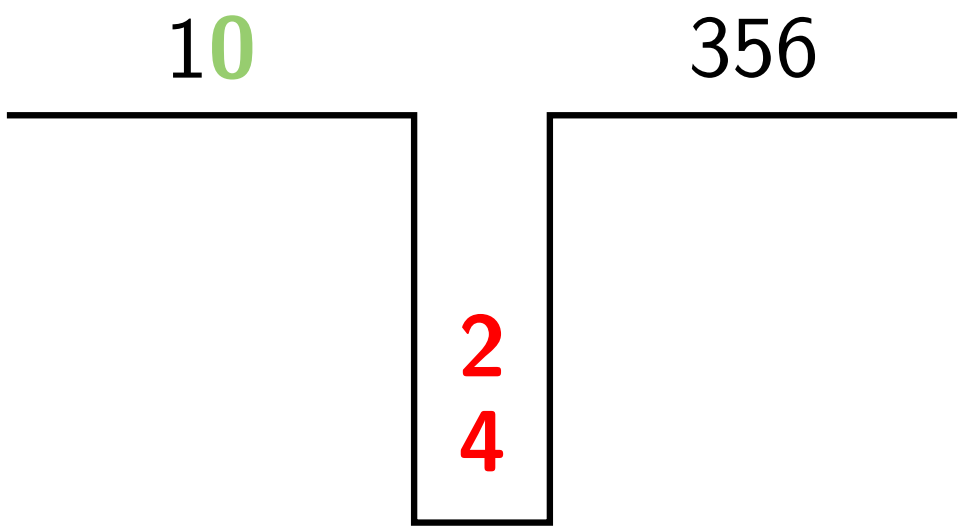


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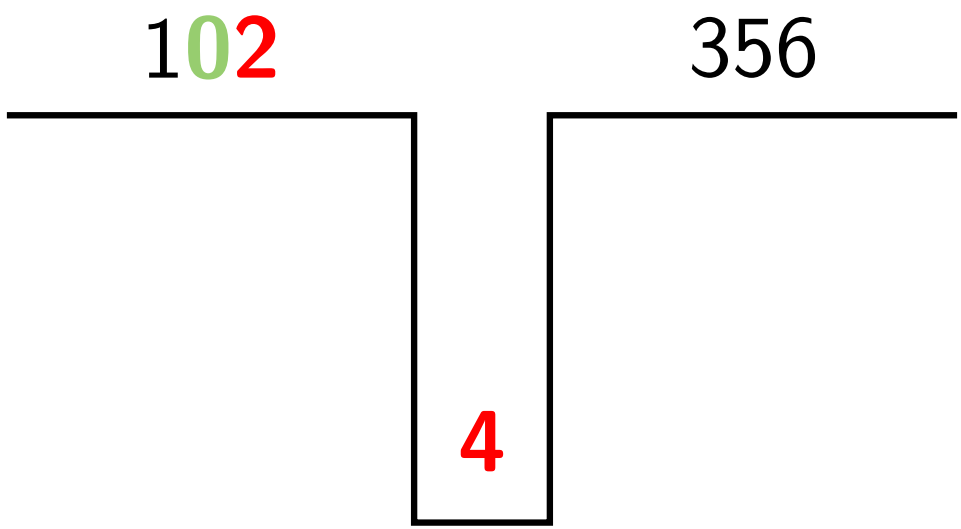


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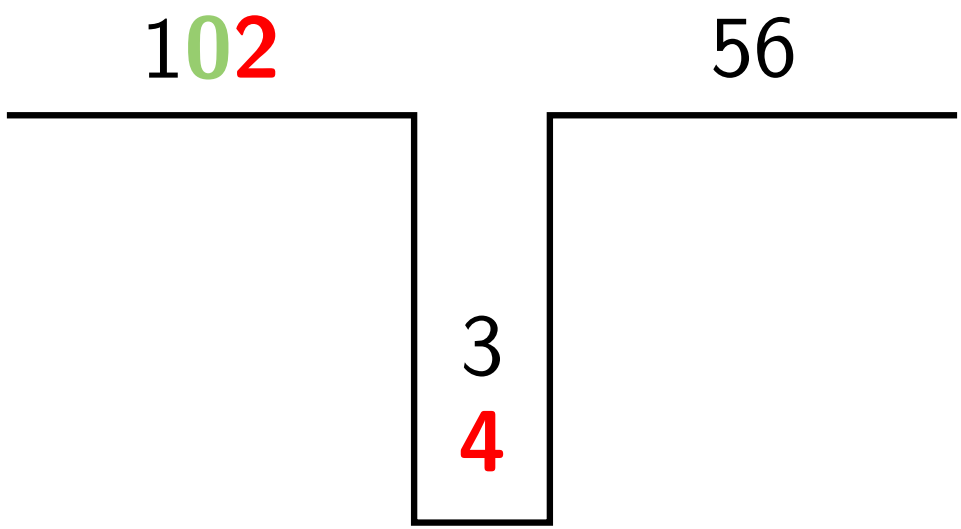


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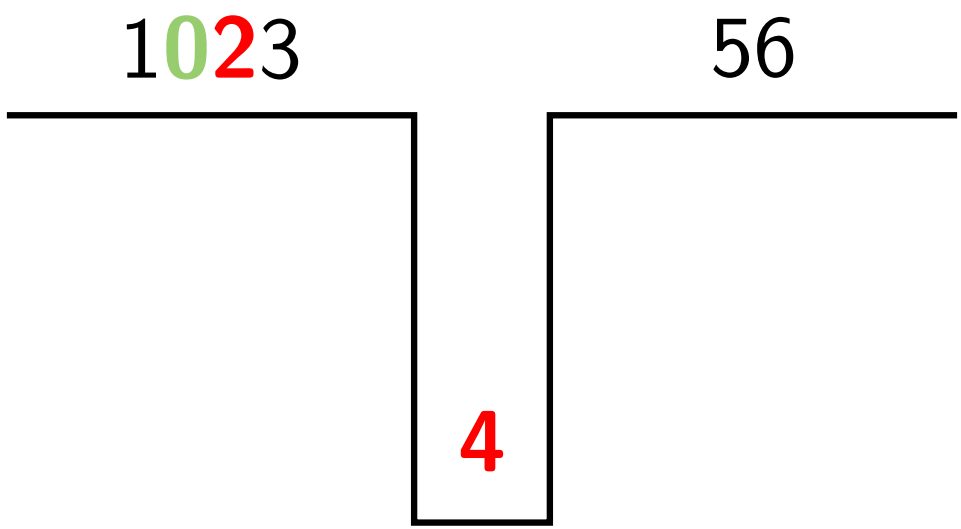


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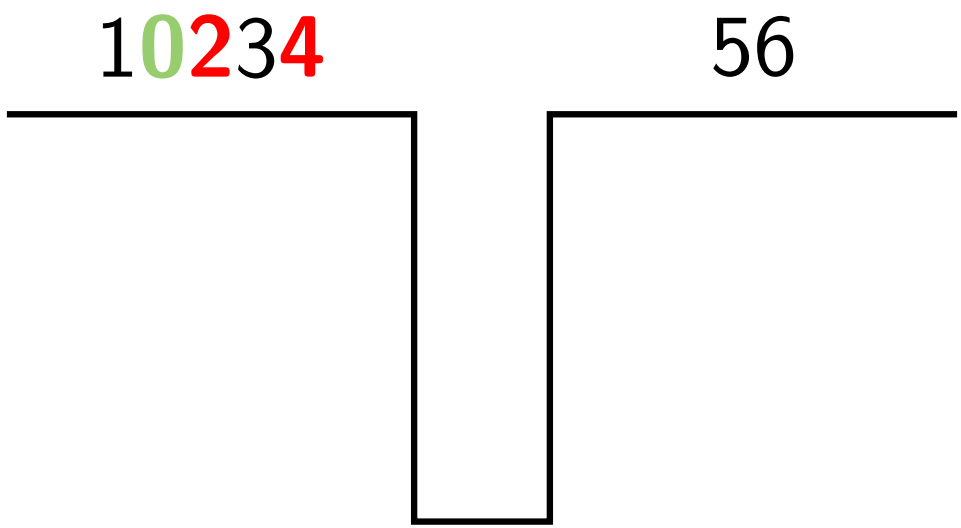


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$$4120356 \overset{s}{\mapsto} 1023456$$

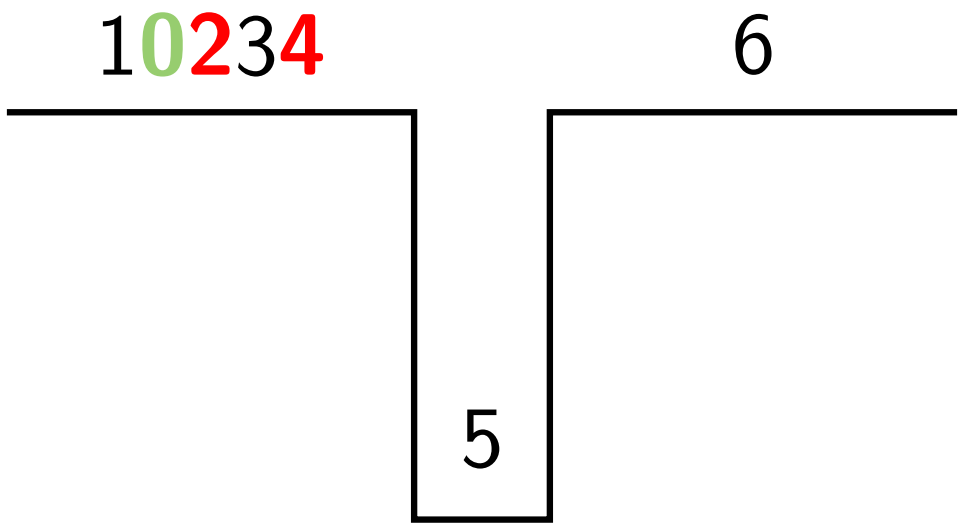


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$$\mathbf{4}1\mathbf{2}\mathbf{0}356 \overset{s}{\mapsto} 1\mathbf{0}\mathbf{2}3\mathbf{4}56$$

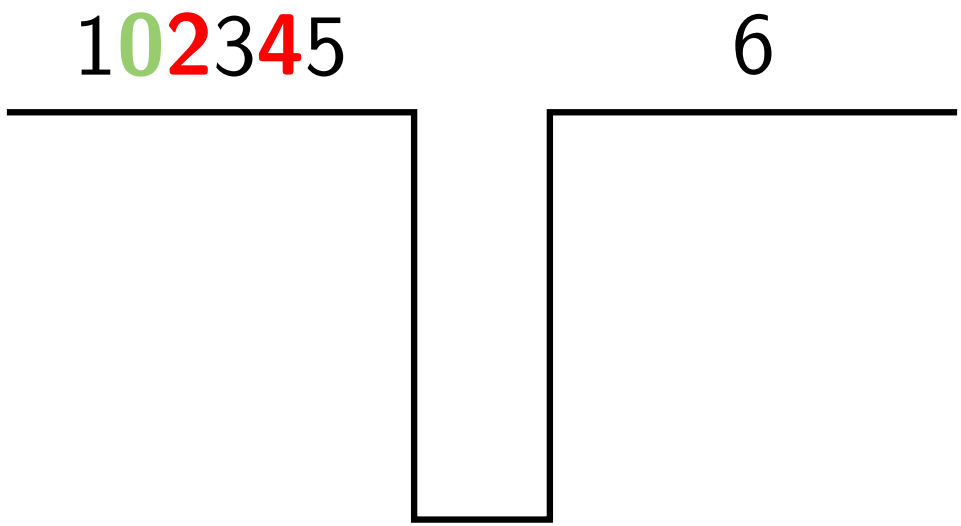


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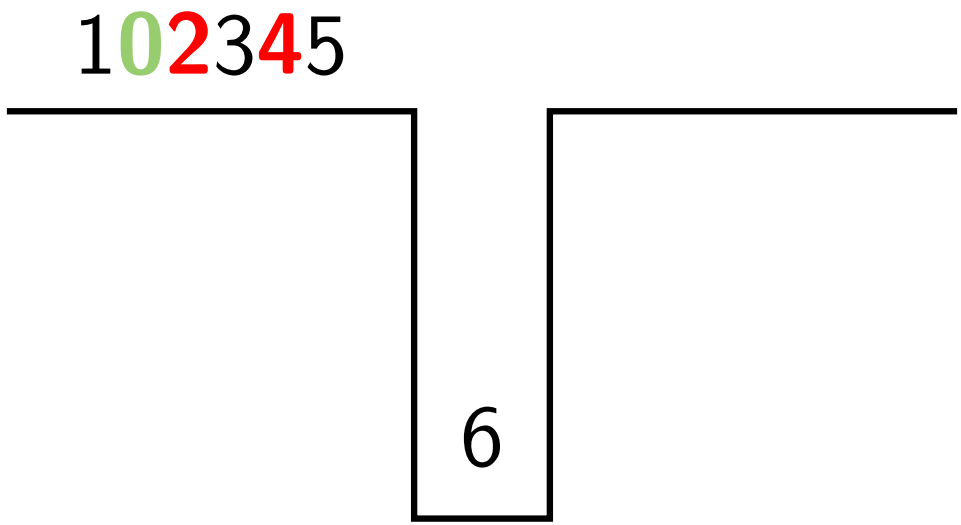


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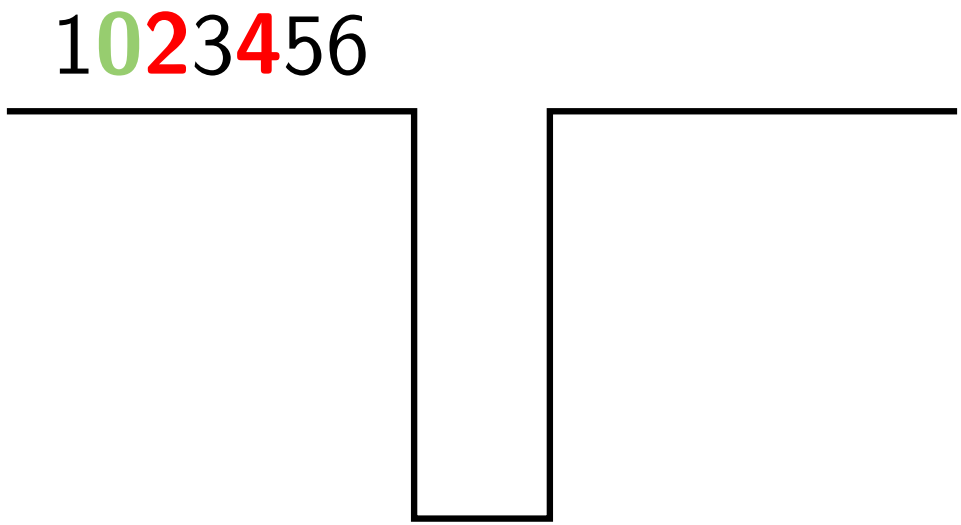


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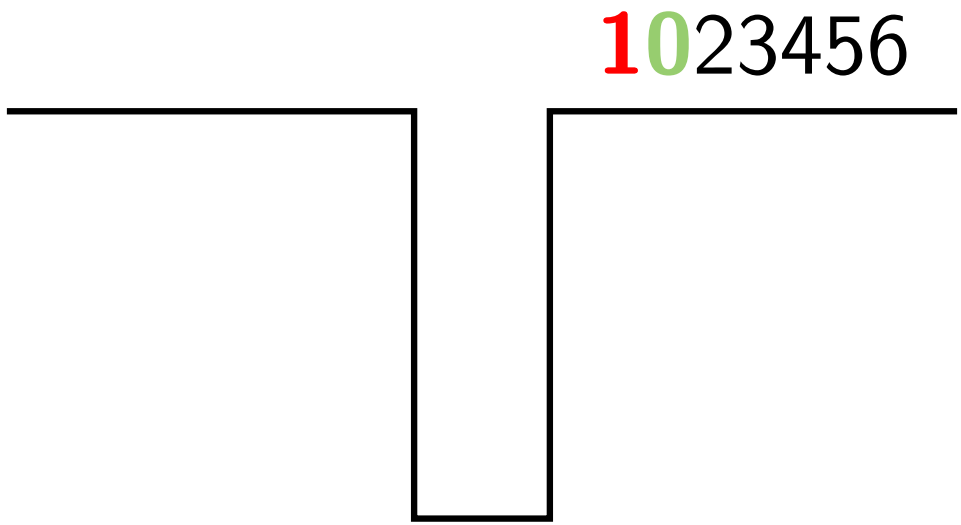


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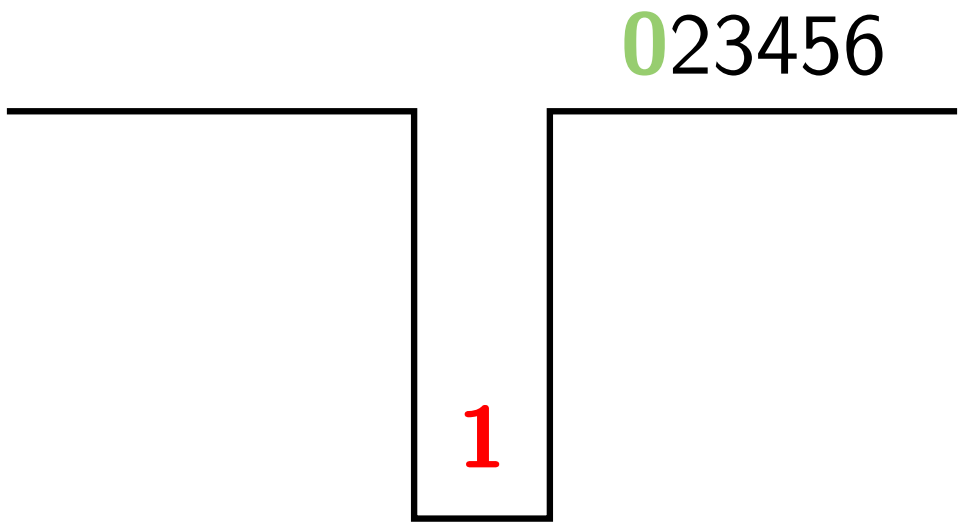


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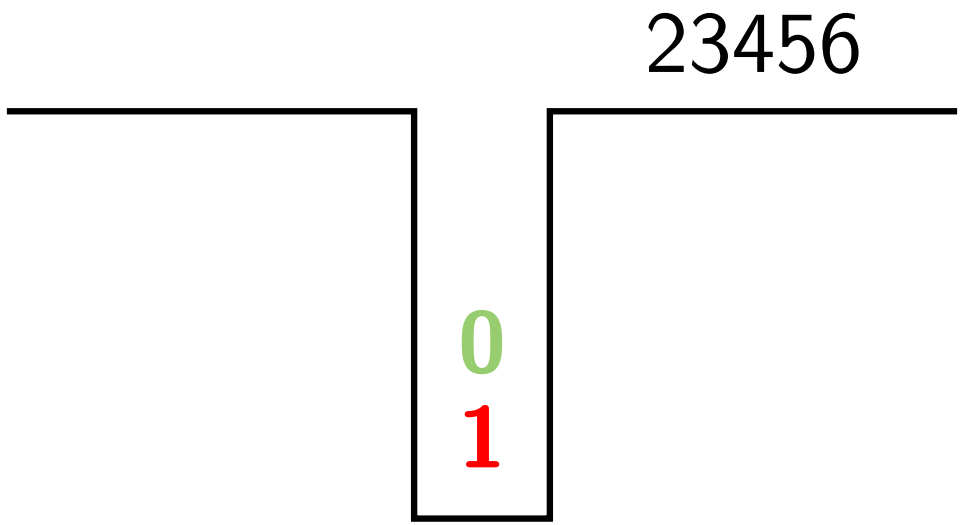


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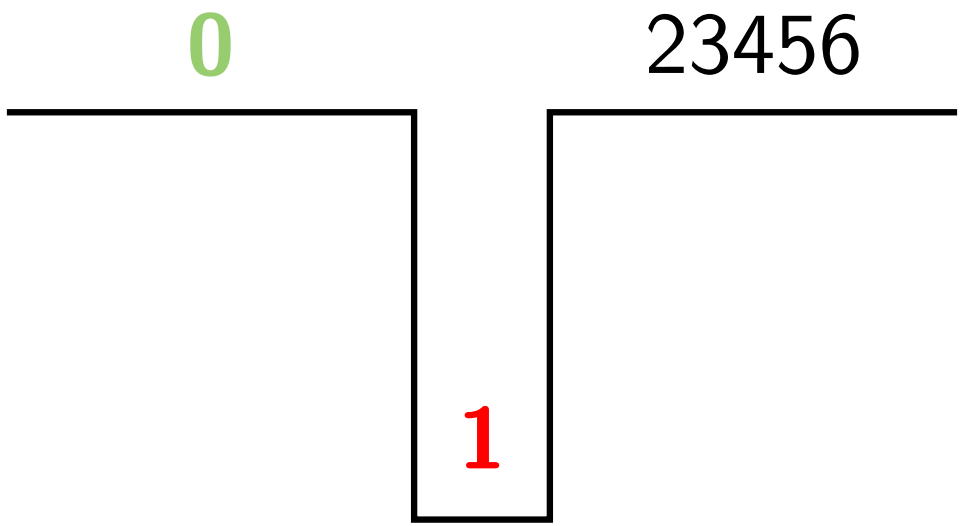


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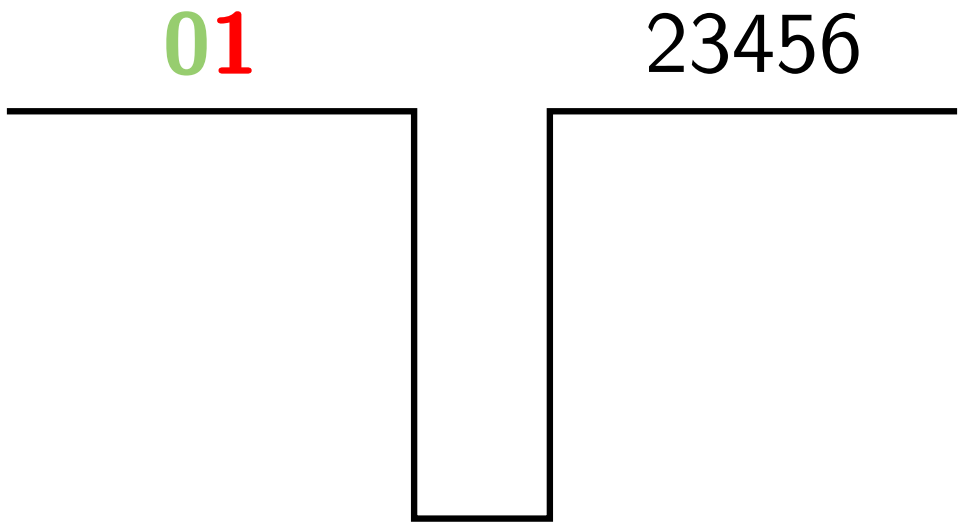


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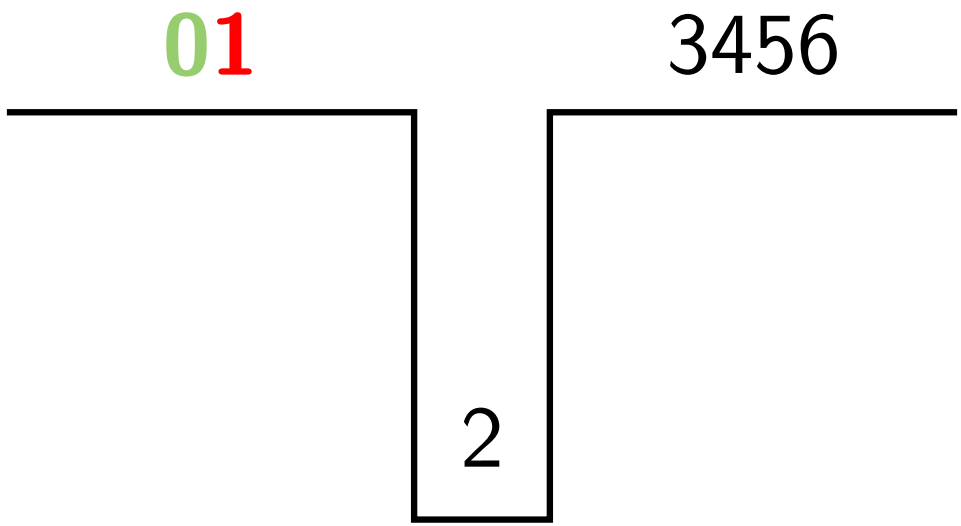


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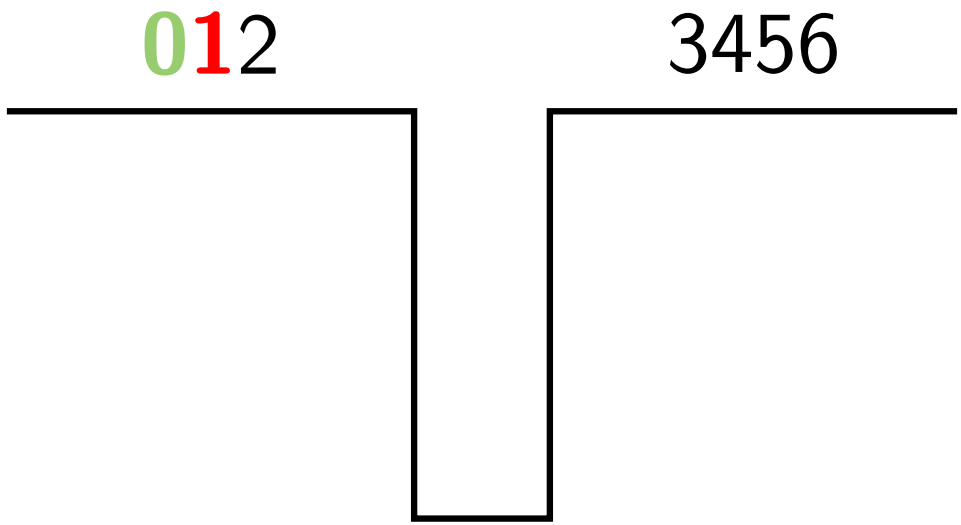


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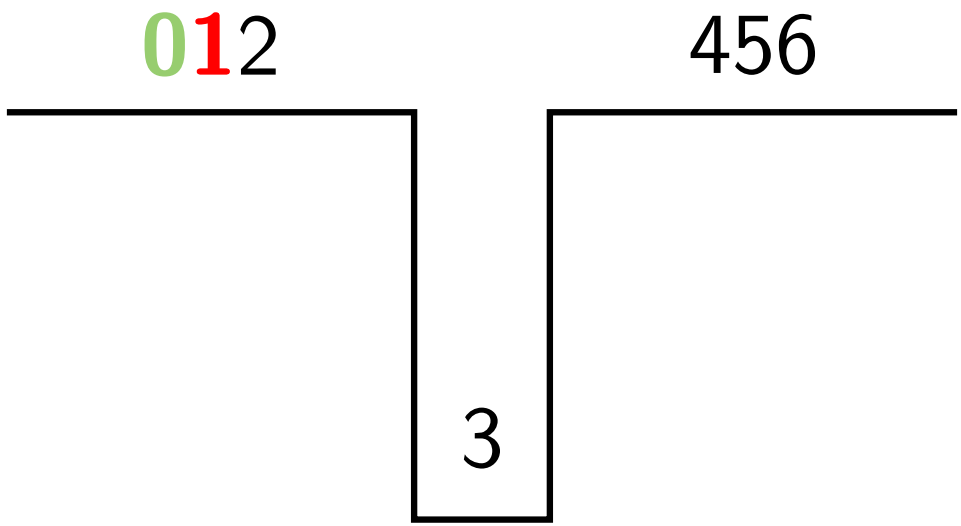


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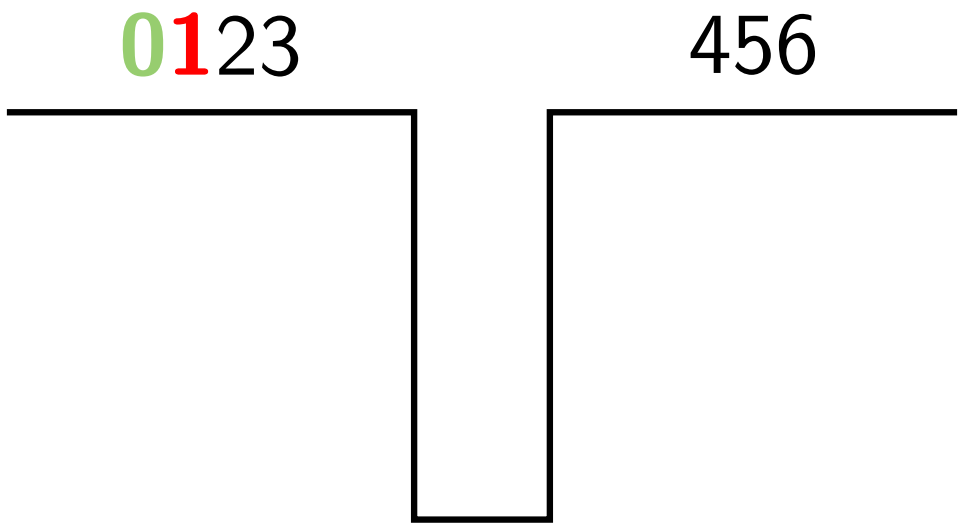


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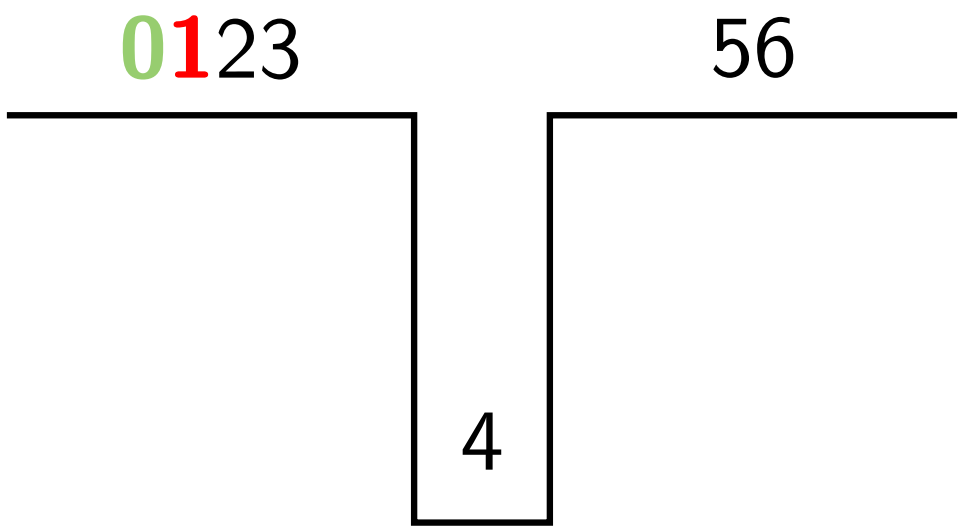


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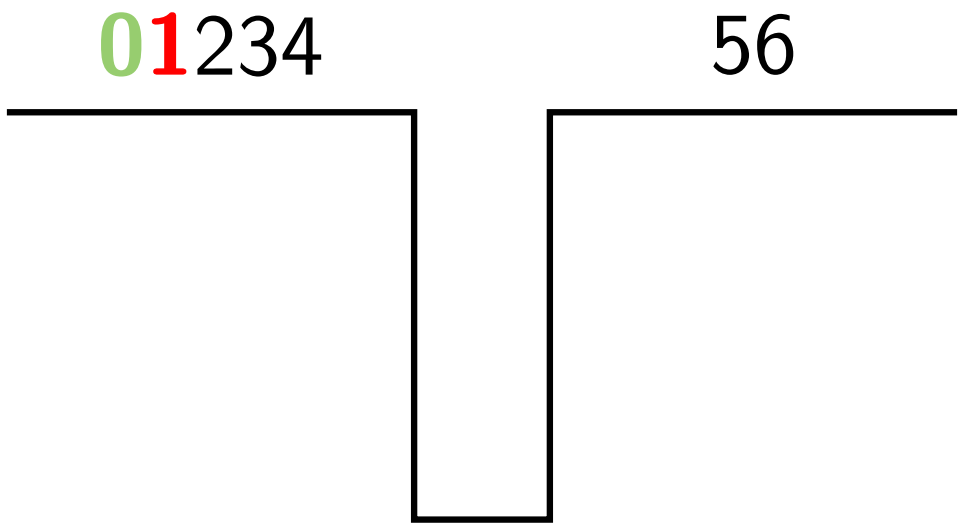


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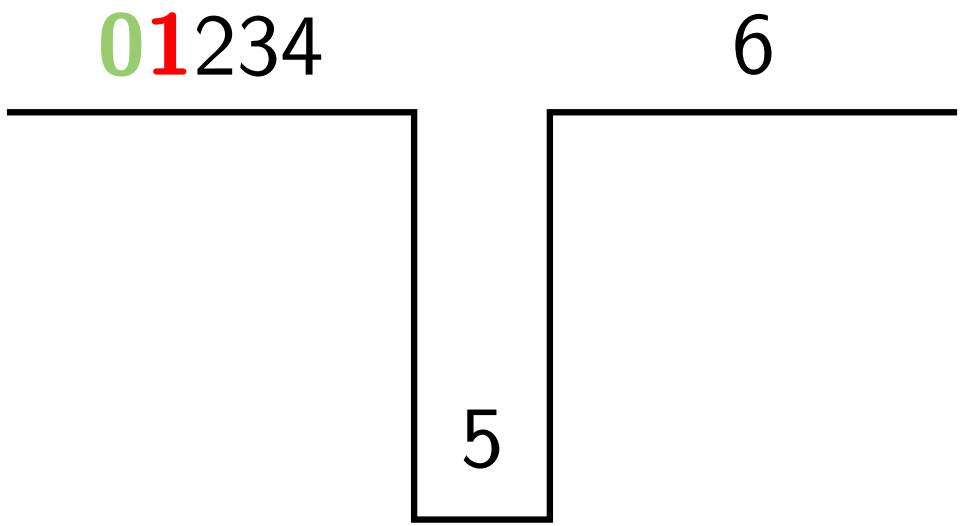


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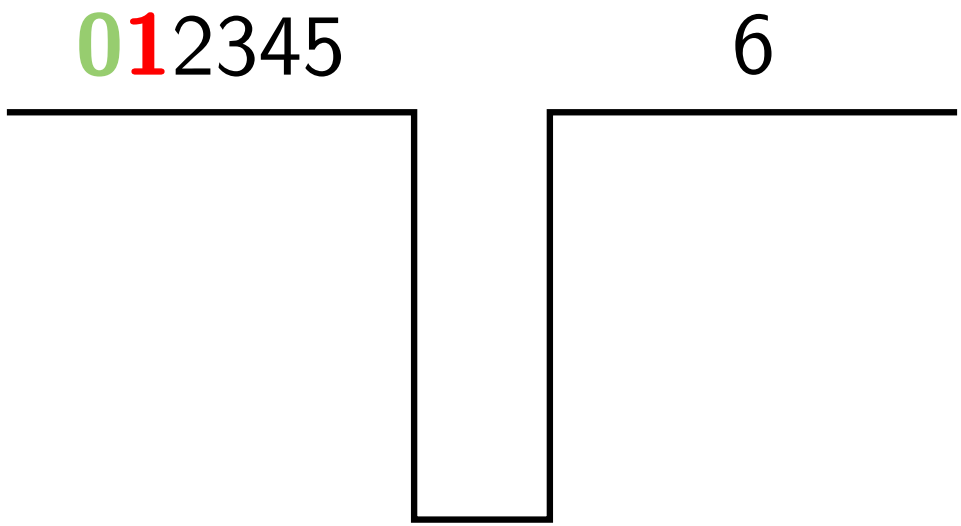


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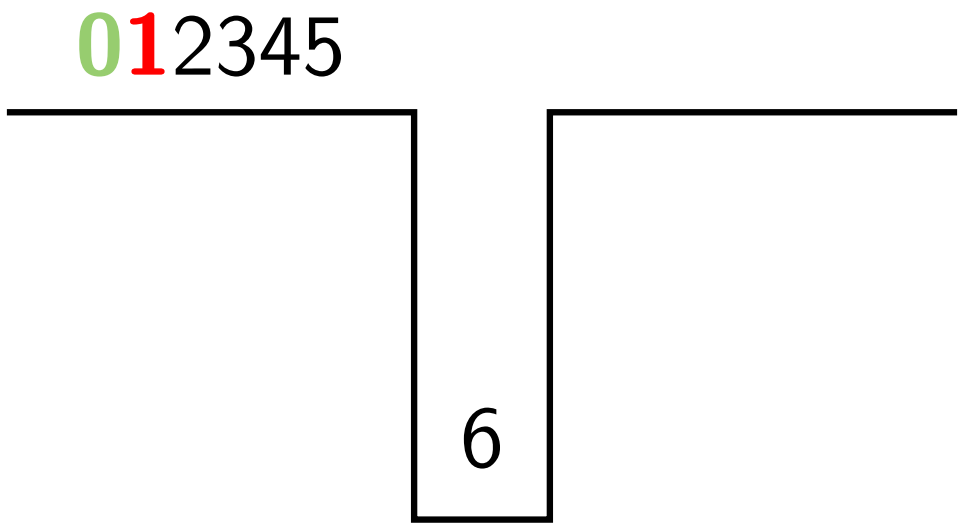


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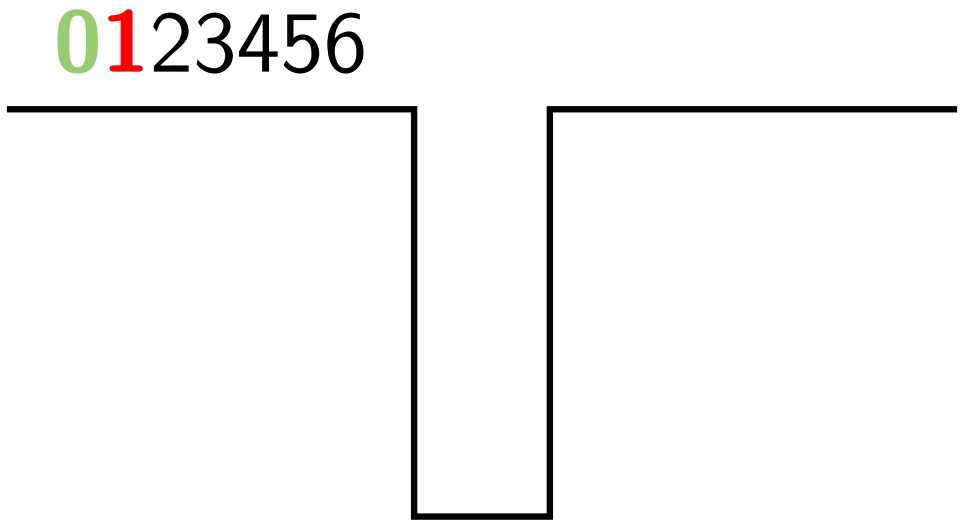


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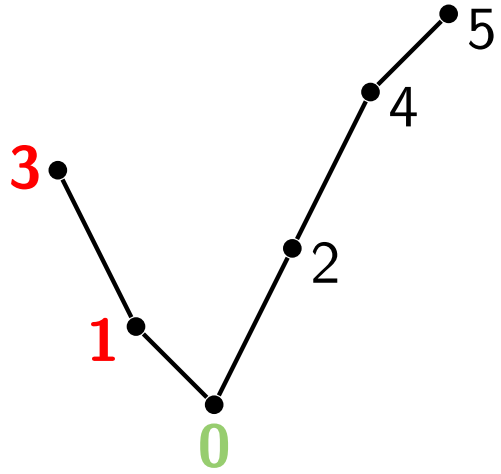
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$$4\mathbf{65}21\mathbf{3}\mathbf{0} \overset{s}{\mapsto} 41\mathbf{2}\mathbf{0}\mathbf{356}$$

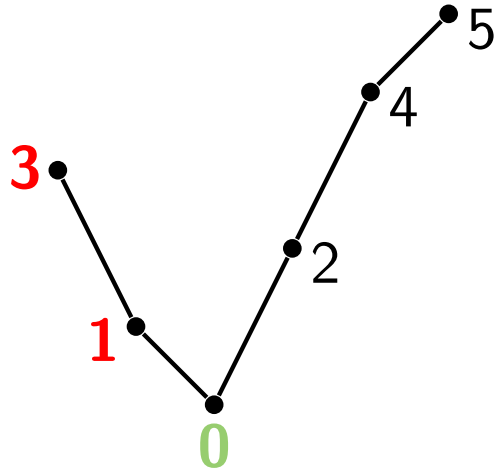
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31**0**245 is s-sortable since all elements before 0 are right-to-left maxima

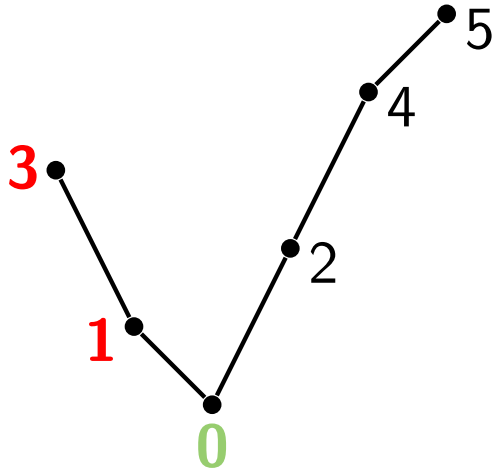


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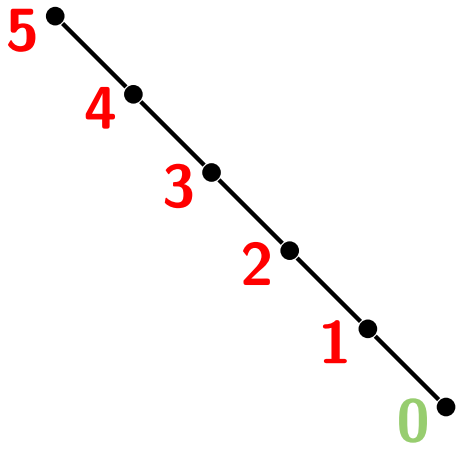


s-sortable permutations are V-shaped!

310245 is s-sortable since all elements before 0 are right-to-left maxima



s-sortable permutations are V-shaped! Only one s-sortable in S'_n:



If $\pi \in S_n'$ is s^2-sortable, ≤ 1 elements between RL-maxima of π



good



bad

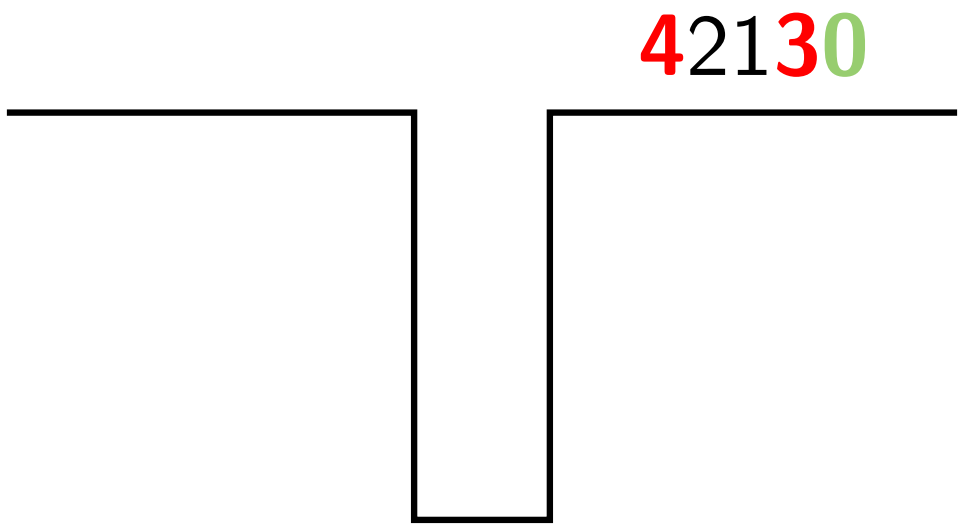
If $\pi \in S_n'$ is s^2-sortable, ≤ 1 elements between RL-maxima of π

42310

good

42130

bad



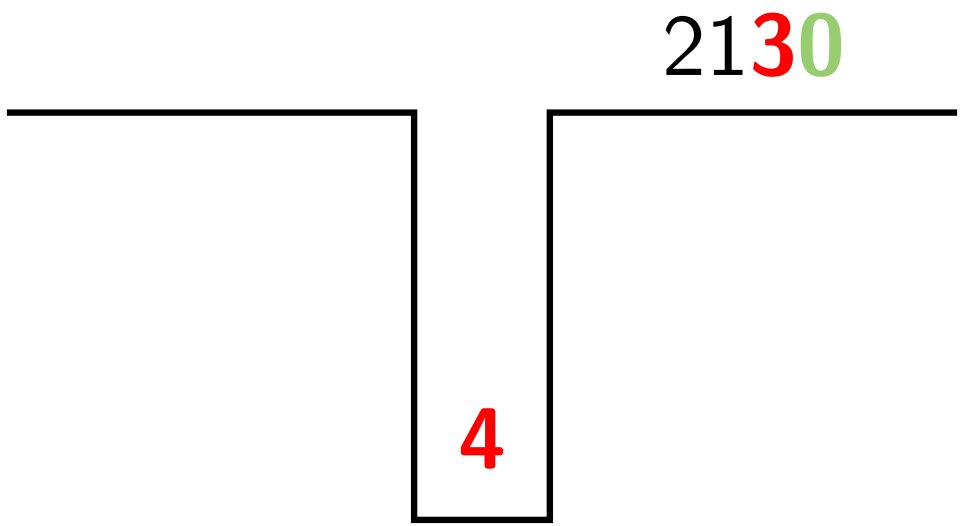
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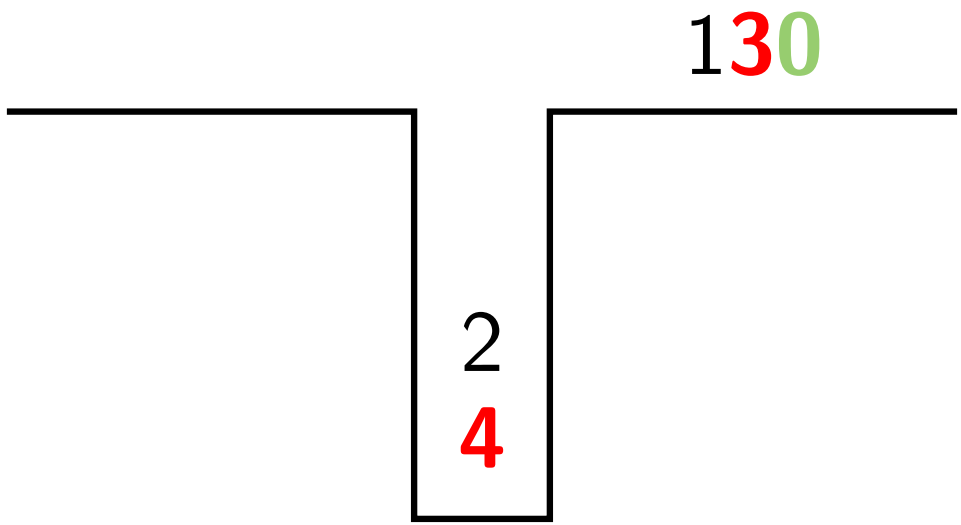
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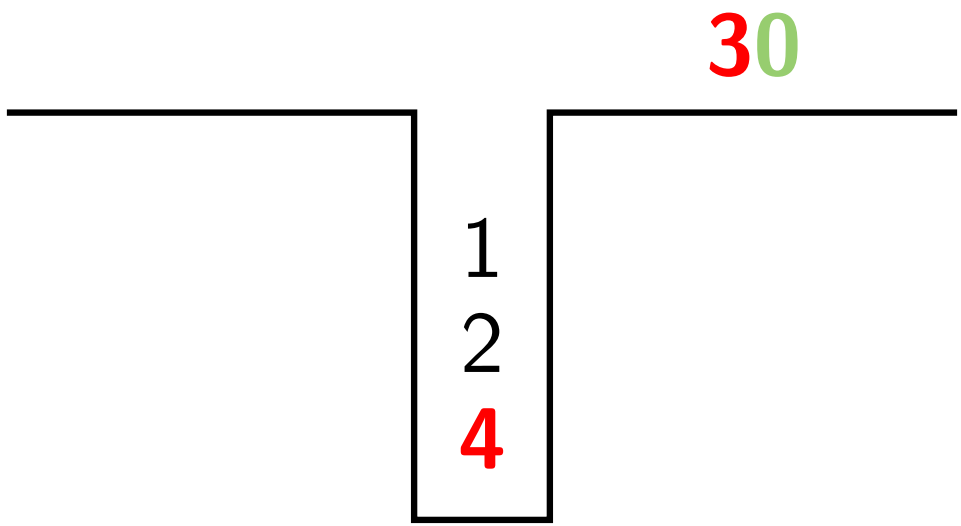
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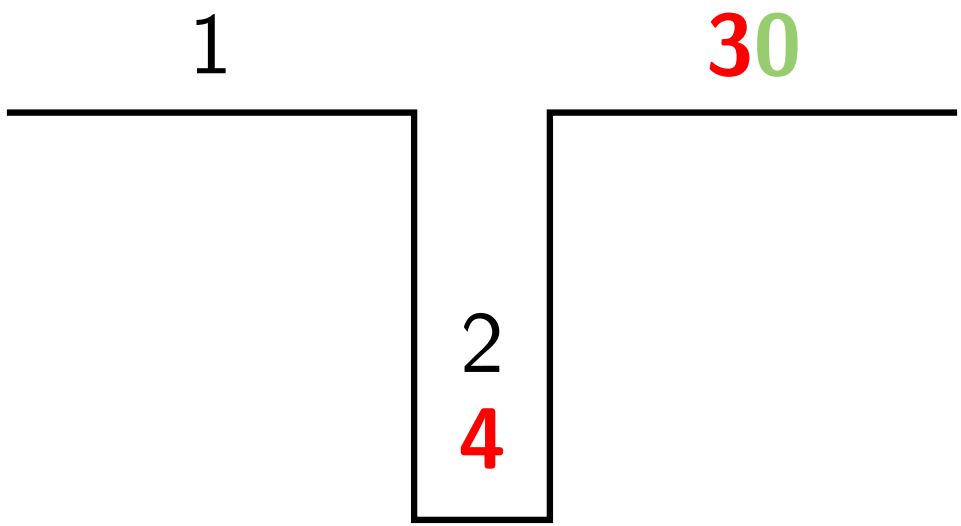
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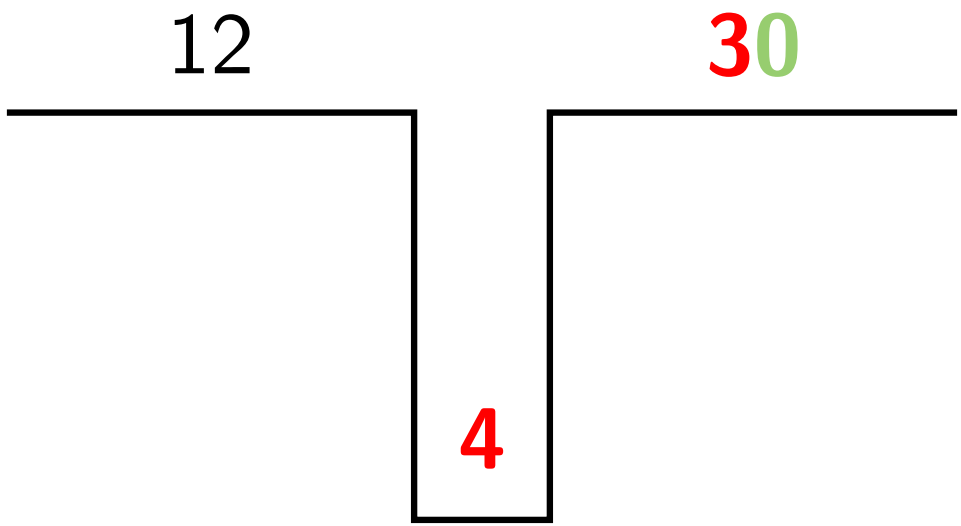
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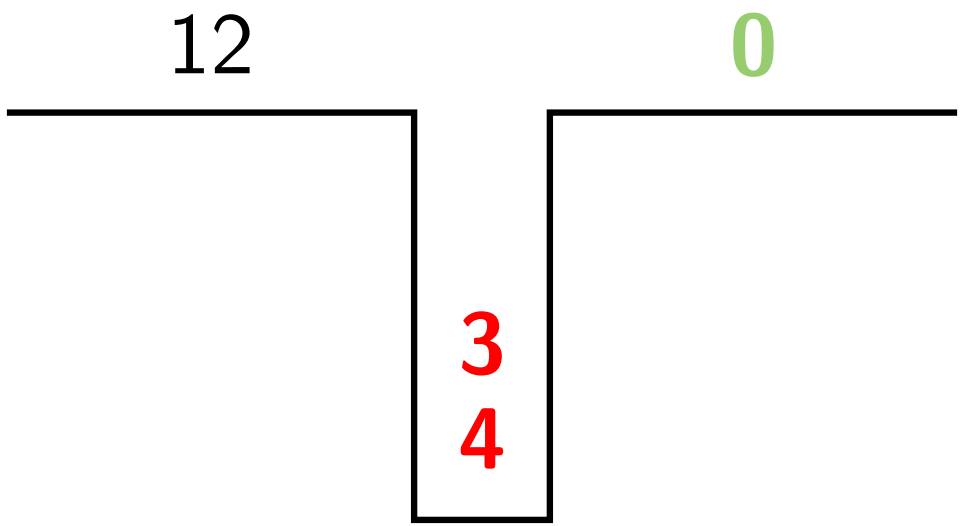
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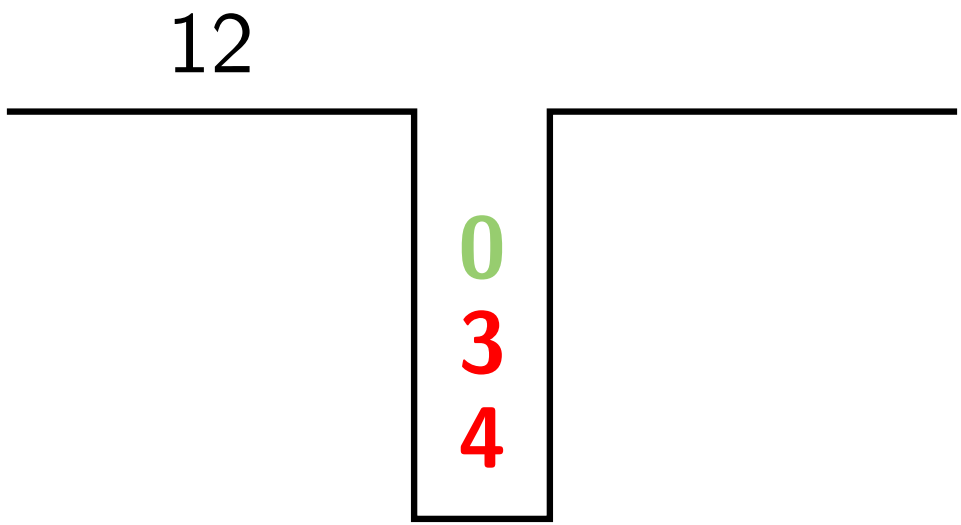
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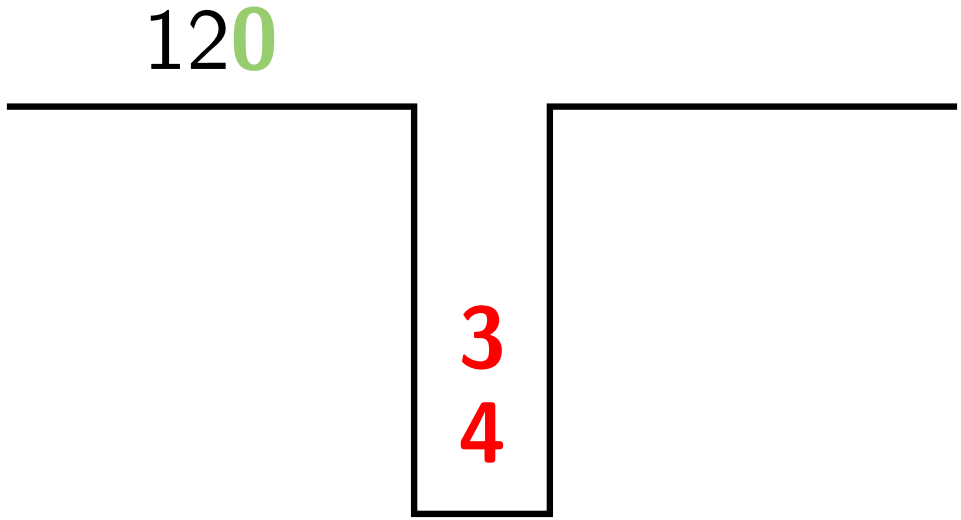
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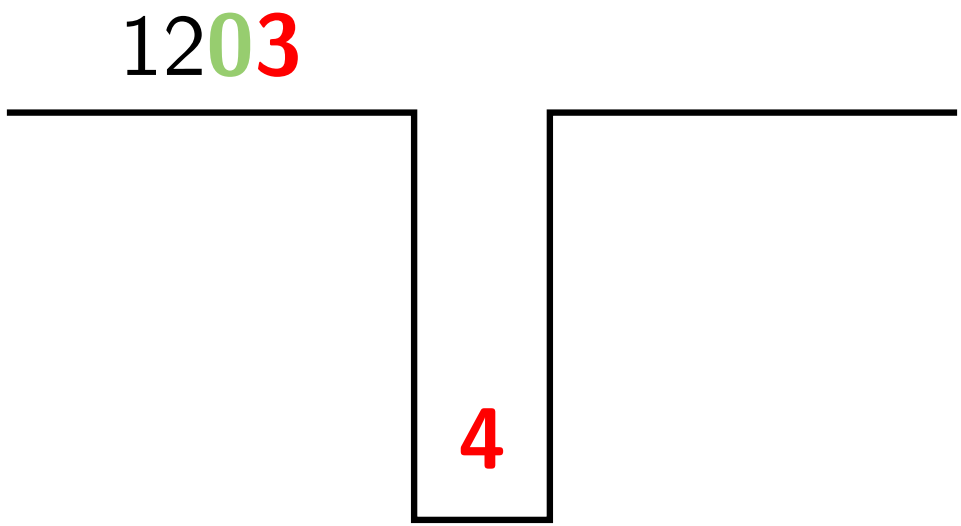
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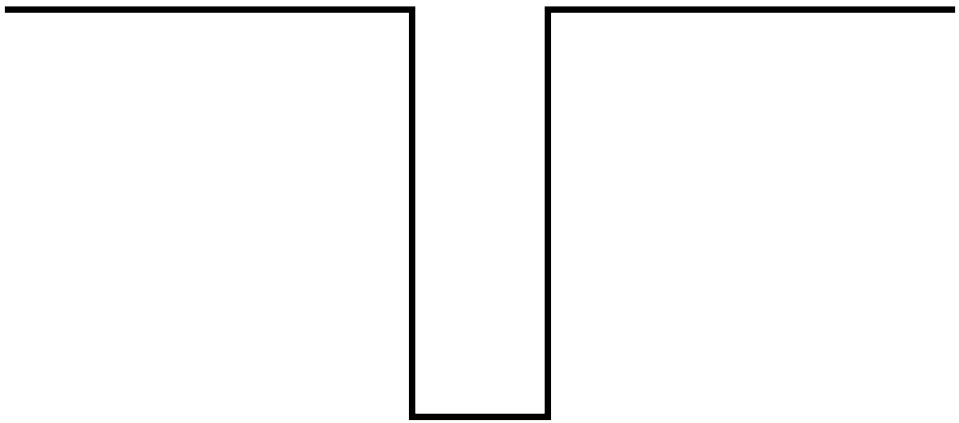
42310

good

42130

bad

12034



354120

4876351 20

$$354120 \longleftrightarrow$$

5	3
4	
2	1

$$48763512 0 \longleftrightarrow$$

8	4
7	
6	
5	3
2	1

$$354120 \longleftrightarrow$$

5	3
4	
2	1

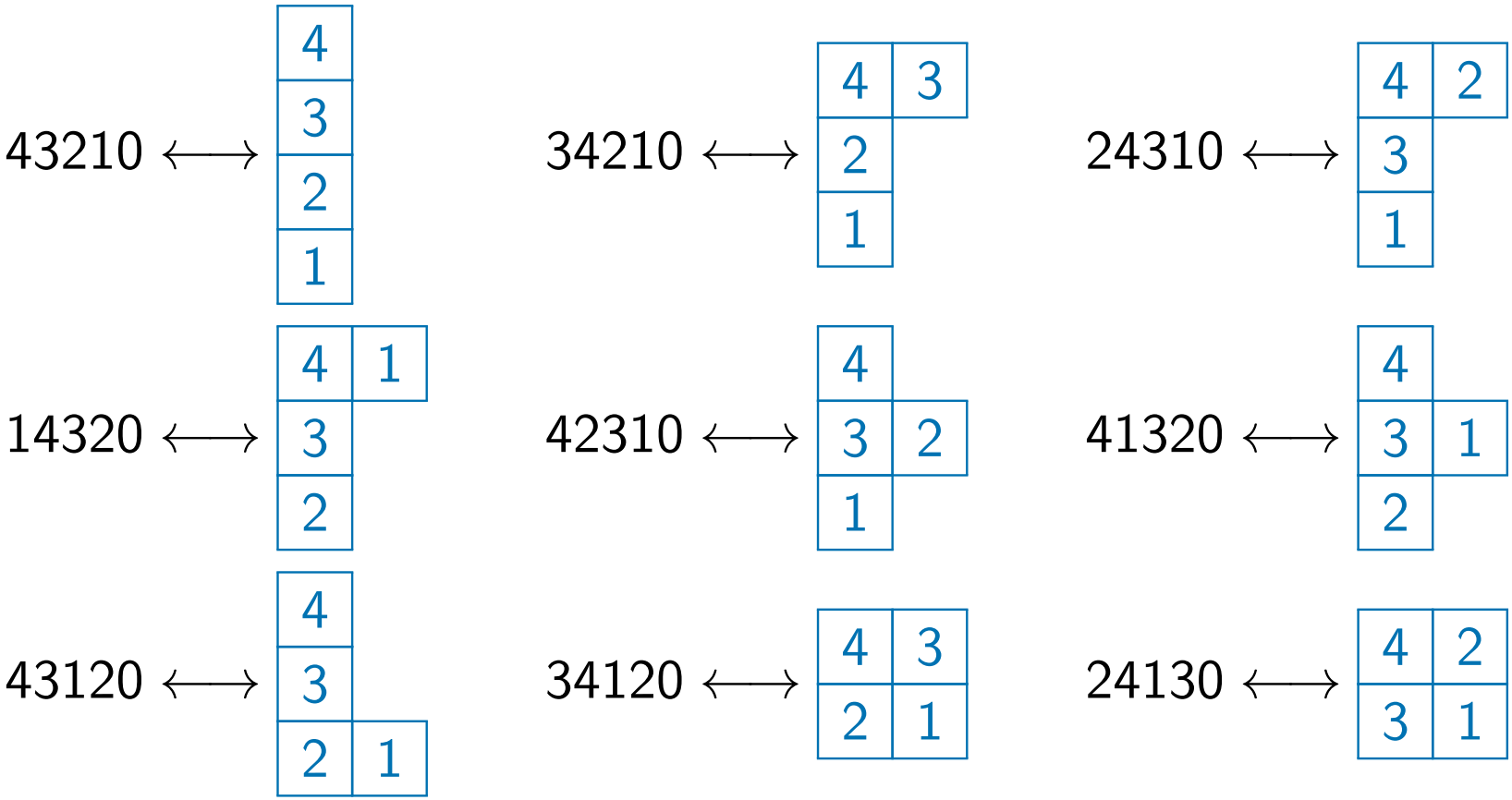
$$48763512 0 \longleftrightarrow$$

8	4
7	
6	
5	3
2	1

- Rows and columns decreasing
- Entries are exactly $\{1, 2, \ldots, n\}$
- Width at most 2

"Antistandard composition tableaux with width at most 2"

s^2-sortable of $S'_n \longleftrightarrow$ antistandard composition tableaux, width ≤ 2



My result:

antistandard composition tableaux, width ≤ 2 $\longleftrightarrow$

standard Young tableaux, width ≤ 3

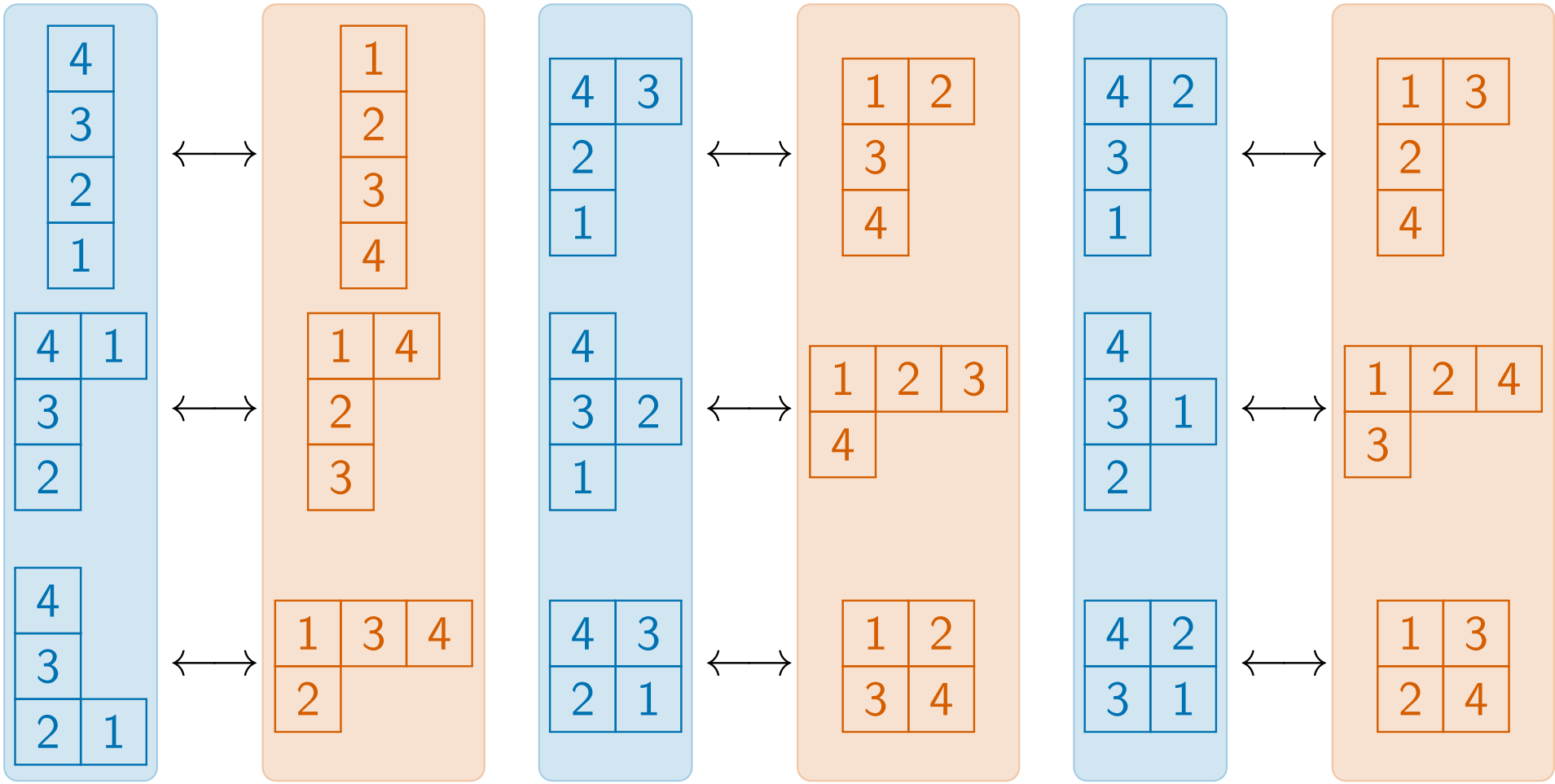


Figure generated with AI assistance.

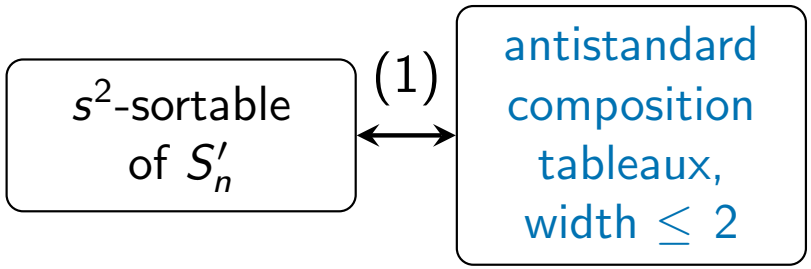


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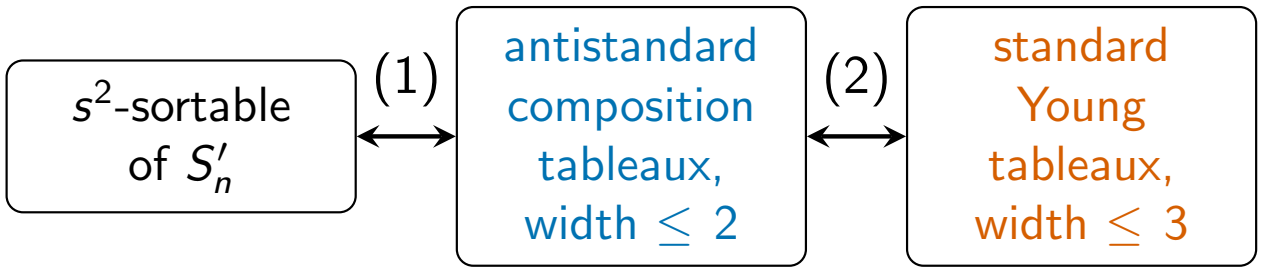


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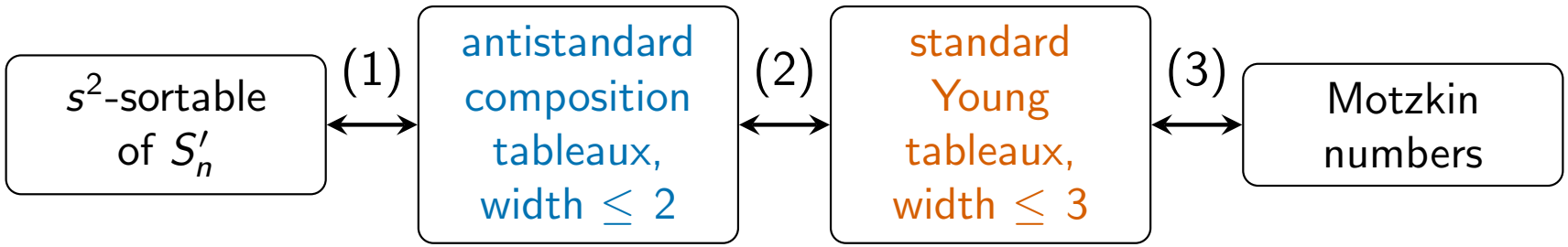
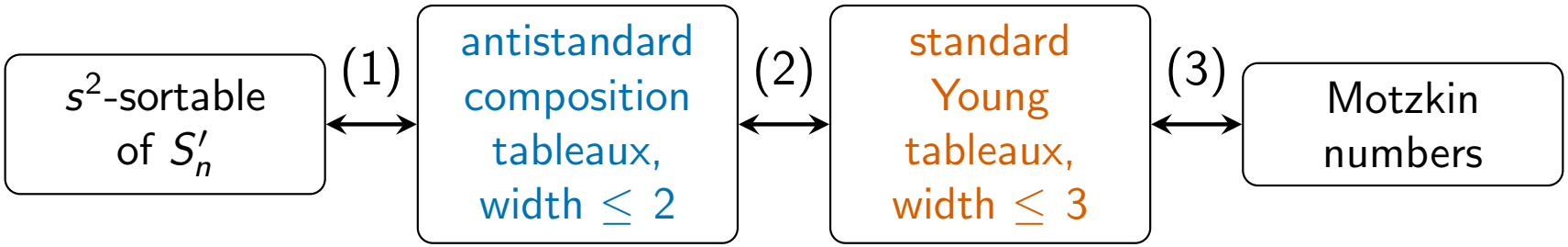


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Theorem (L. 2026)

of s^2-sortable permutations in S'_n is nth Motzkin number.

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*n*th Motzkin counts:

Number of 4321−, (3412,2413)−, (3412,3142)− and 3412−avoiding involutions in S_n.
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The Hankel transform of this sequence with a(0) omitted gives the period−6 sequence [1, 0, −1, −1, 0, 1, ...] which is A010892 with its first term omitted, while the Hankel transform of the current sequence is the all−ones sequence A000012, and also it is the unique sequence with this property which is similar to the unique Hankel transform property of the Catalan numbers. − Michael Somos, Apr 17 2022
The number of terms in which the exponent of any variable x_i is not greater than 2 in the expansion of Product_{j=1..n} Sum_{i=1..j} x_i. E.g.: a(4) = 9: 3*x1^2*x2^2, 4*x1^2*x2*x3, 2*x1^2*x2*x4, x1^2*x3^2, x1^2*x3*x4, 2*x1*x2^2*x3, x1*x2^2*x4, x1*x2*x3^2, x1*x2*x3*x4. − Elif Baser, Dec 20 2024
The Hankel matrix corresponding to this sequence factorizes as T * Transpose(T), where T is the lower unitriangular array A064189 (row reverse of the Motzkin triangle A026300). Hence the Hankel determinants are all equal to 1 as stated above by Michael Somos. − Peter Bala, Nov 23 2025
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(2) For prime p == 5 (mod 6) and n >= 1, a(n*p − 2) == −A005773(n) (mod p).
(3) For prime p >= 3 and k >= 1, a(n + p^k) == a(n) (mod p) for 0 <= n <= (p^k − 3).
(4) For prime p >= 5 and k >= 2, a(n + p^k) == a(n) (mod p^2) for 0 <= n <= (p^(k−1) − 3). (End)
The Hankel transform of this sequence with a(0) omitted gives the period−6 sequence [1, 0, −1, −1, 0, 1, ...] which is A010892 with its first term omitted, while the Hankel transform of the current sequence is the all−ones sequence A000012, and also it is the unique sequence with this property which is similar to the unique Hankel transform property of the Catalan numbers. − Michael Somos, Apr 17 2022
The number of terms in which the exponent of any variable x_i is not greater than 2 in the expansion of Product_{j=1..n} Sum_{i=1..j} x_i. E.g.: a(4) = 9: 3*x1^2*x2^2, 4*x1^2*x2*x3, 2*x1^2*x2*x4, x1^2*x3^2, x1^2*x3*x4, 2*x1*x2^2*x3, x1*x2^2*x4, x1*x2*x3^2, x1*x2*x3*x4. − Elif Baser, Dec 20 2024
The Hankel matrix corresponding to this sequence factorizes as T * Transpose(T), where T is the lower unitriangular array A064189 (row reverse of the Motzkin triangle A026300). Hence the Hankel determinants are all equal to 1 as stated above by Michael Somos. − Peter Bala, Nov 23 2025
From Michael Luo, Jul 29 2026: (Start)
Number of 2−stack−sortable permutations of {1,2,...,n + 1} ending with 1.
Number of standard composition tableaux of width at most 2 on n cells. (End)

From Michael Luo, Jul 29 2026: (Start) Number of 2-stack-sortable permutations of {1,2,...,n + 1} ending with 1.

Online Encyclopedia of Integer Sequences, August 5, 2026

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- Jerry Zhang
- MIT and CEE
- my sponsor Joseph Mandarino

s^2-sortable of $S'_n \longleftrightarrow$ antistandard composition tableaux, width ≤ 2

s^2-sortable of $S'_n \longleftrightarrow$ antistandard composition tableaux, width ≤ 2

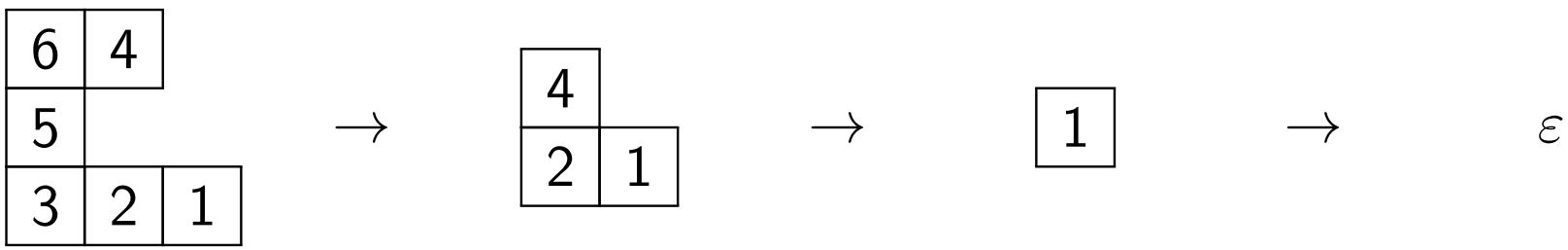
s^t-sortable of $S'_n \longleftrightarrow$ antistandard composition tableaux, width $\leq t$

s^2-sortable of $S'_n \longleftrightarrow$ antistandard composition tableaux, width ≤ 2

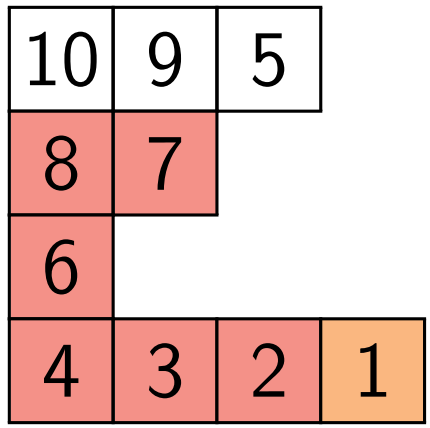
s^t-sortable of $S'_n \longleftrightarrow$ antistandard composition tableaux, width $\leq t$

Applications of s correspond to cutting off columns:

$$4652130 \overset{s}{\mapsto} 4120356 \overset{s}{\mapsto} 1023456 \overset{s}{\mapsto} 0123456$$



Power $p(c)$ of a cell:



Theorem (Zhang 2026)

of words in S'_n *corresponding to* T *is*

$$\prod_c p(c),$$

where product taken over $c \in T$ *not in first column.*

Theorem (Zhang 2026)

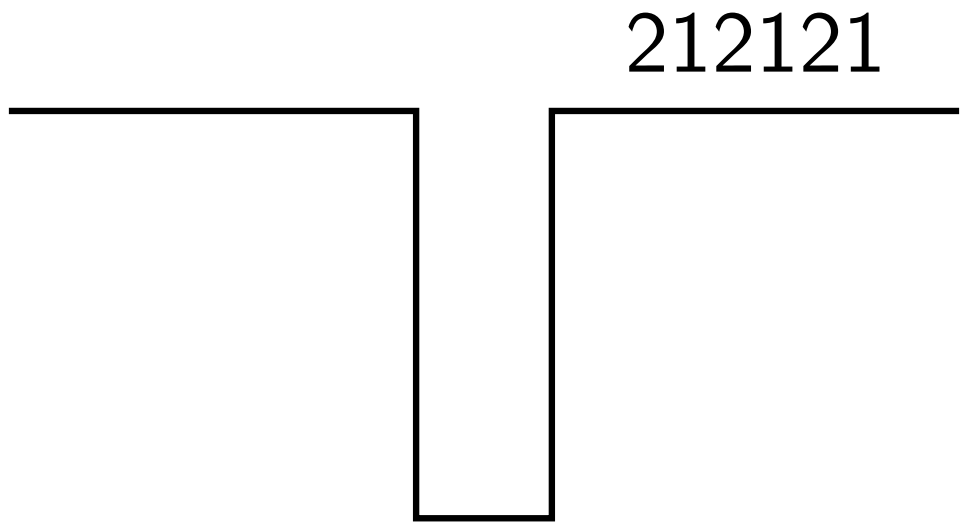
The number of s^t-sortable permutations in S'_n is computable in polynomial time in n.

Theorem (Zhang 2026)

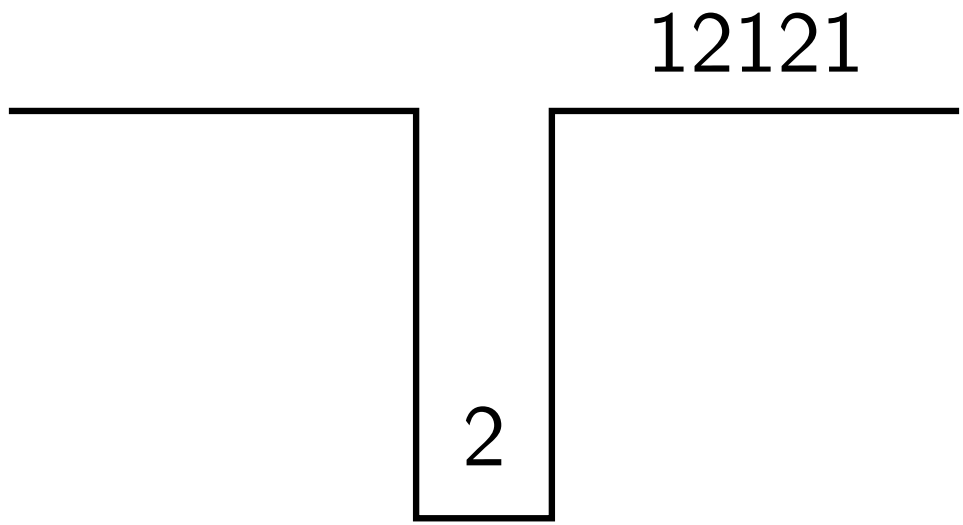
Let $f(n)$ be the average number of iterations needed to sort $\pi \in S_n$. Then $\lim_{n\to\infty} \frac{f(n)}{n}$ exists.

s_m allows m identical penguins in hole.

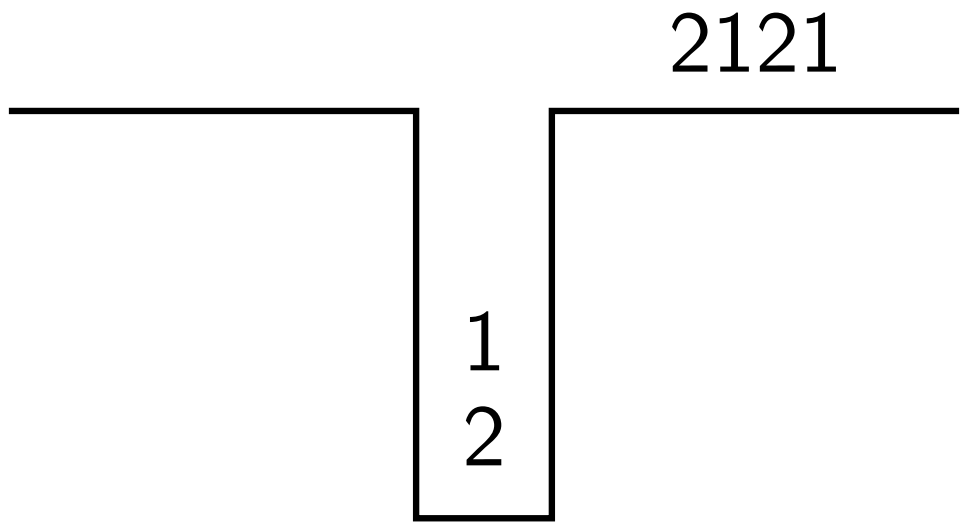
s_m allows m identical penguins in hole. s_2:



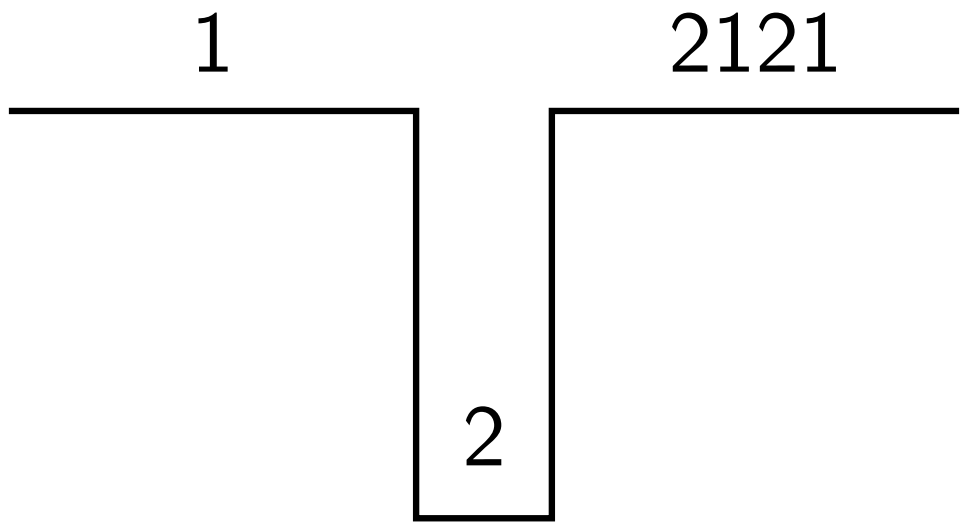
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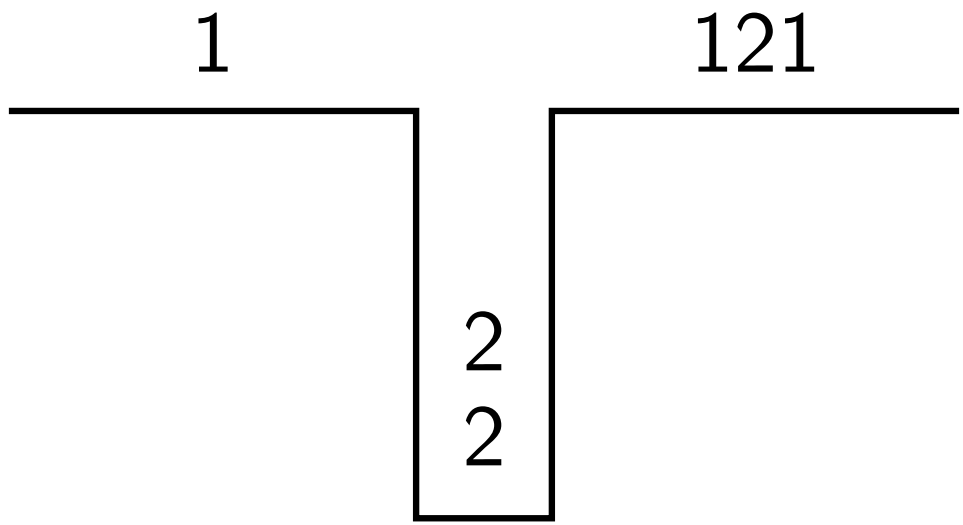
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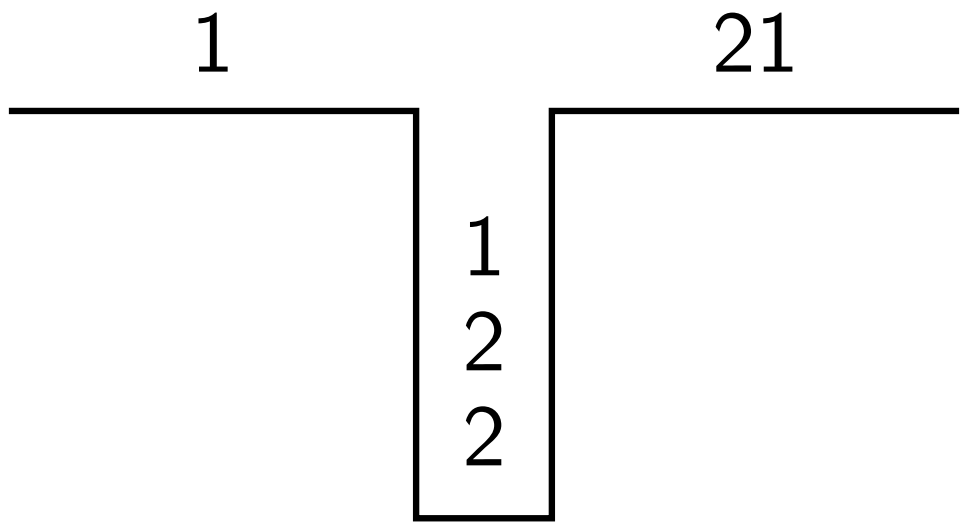
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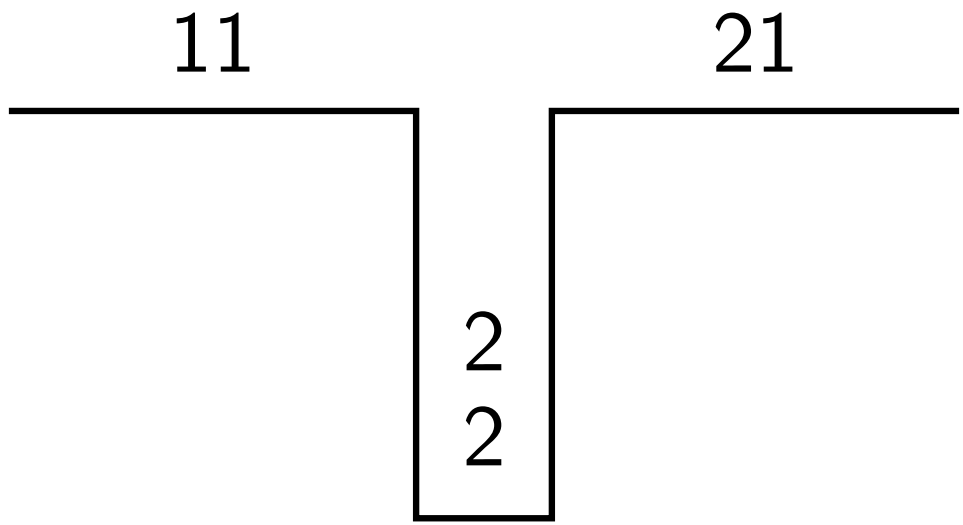
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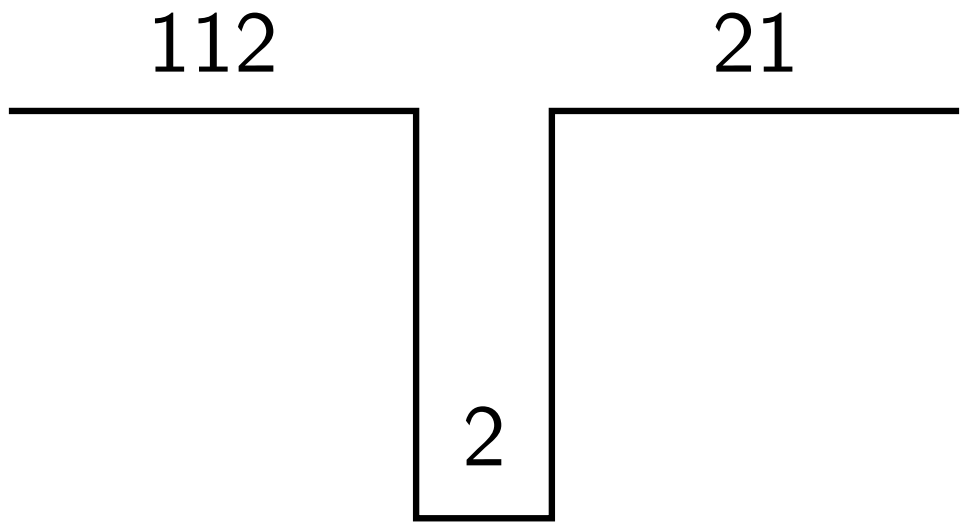
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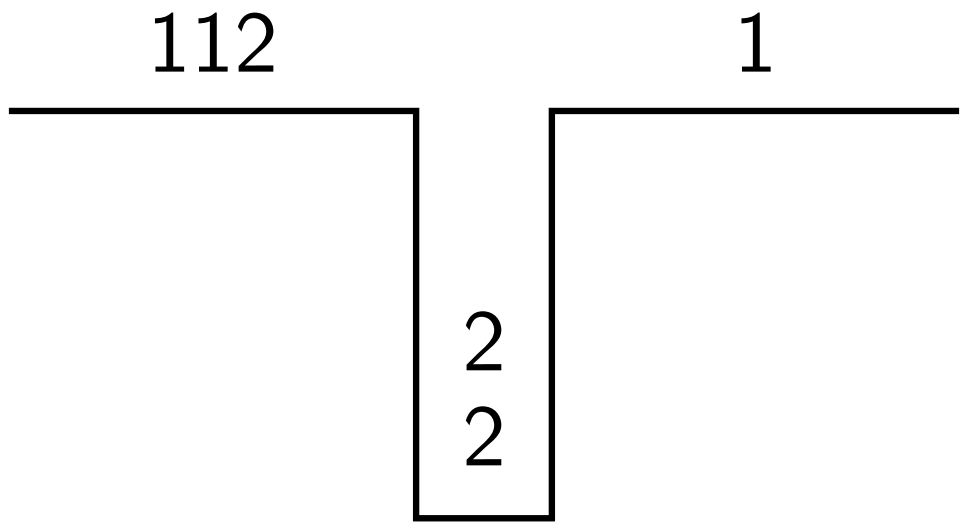
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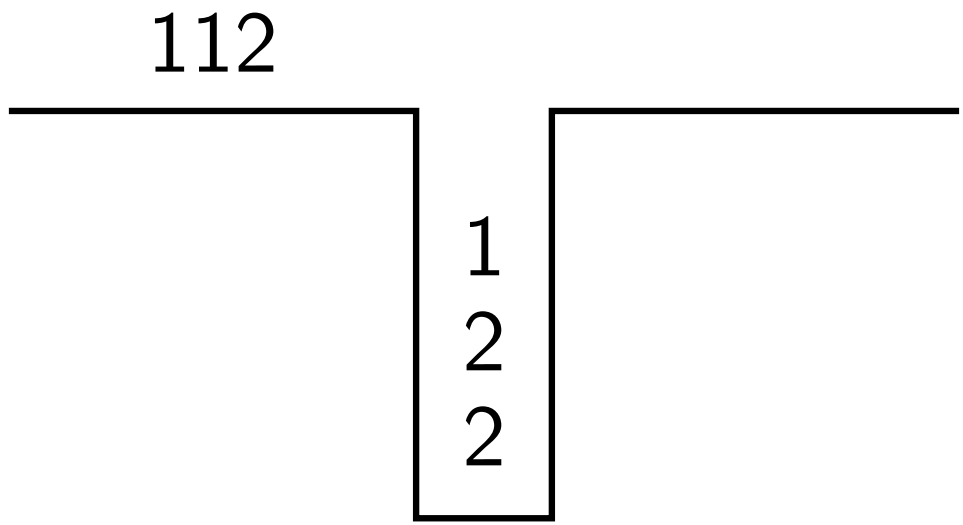
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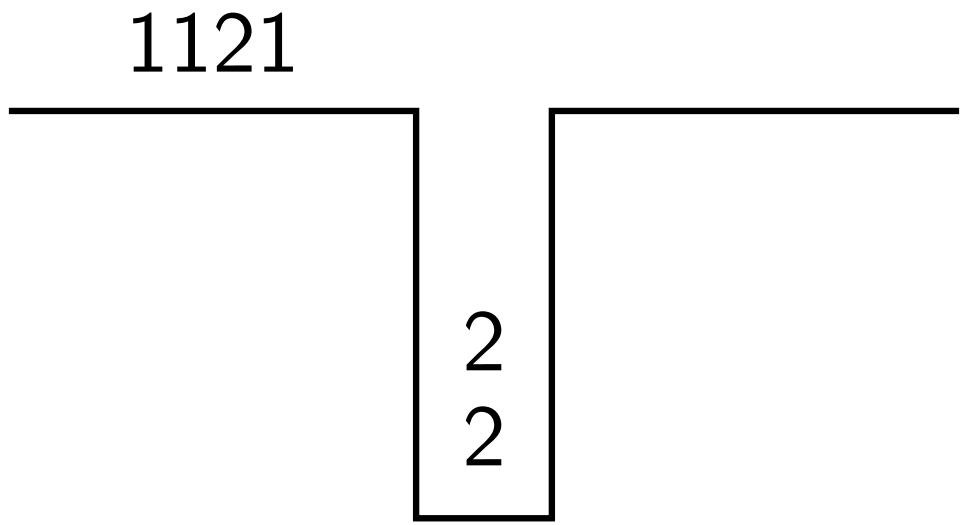
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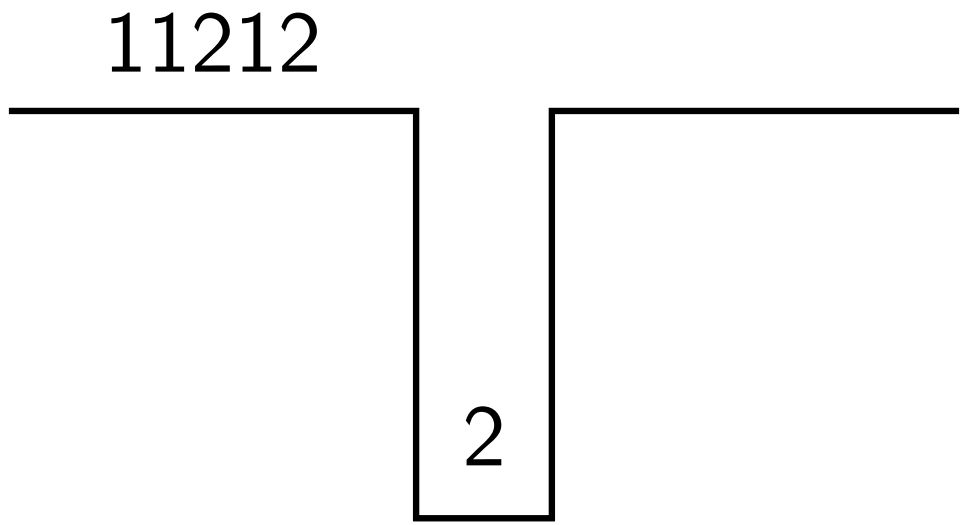
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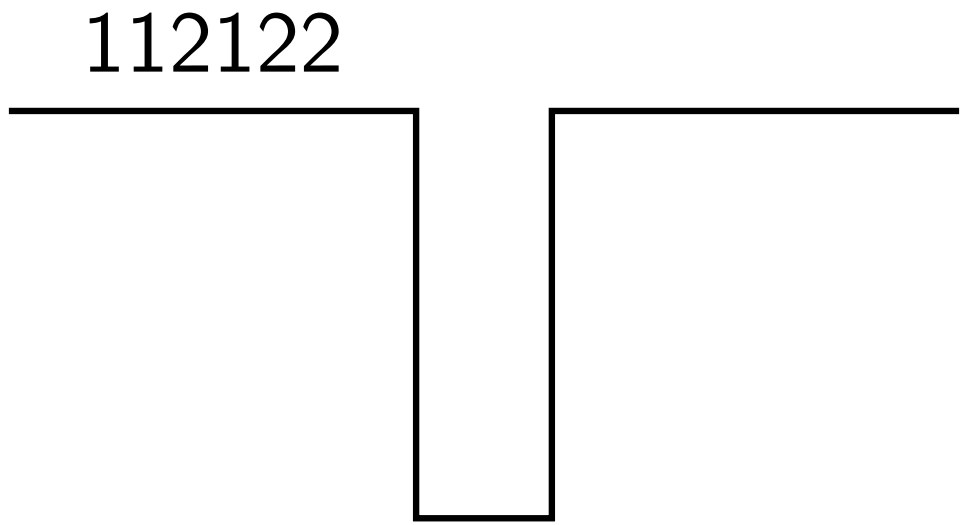
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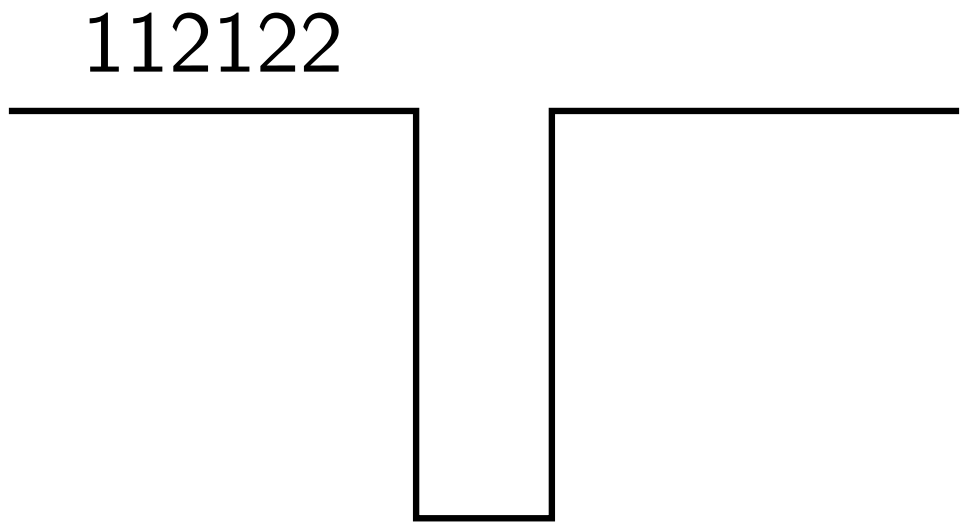
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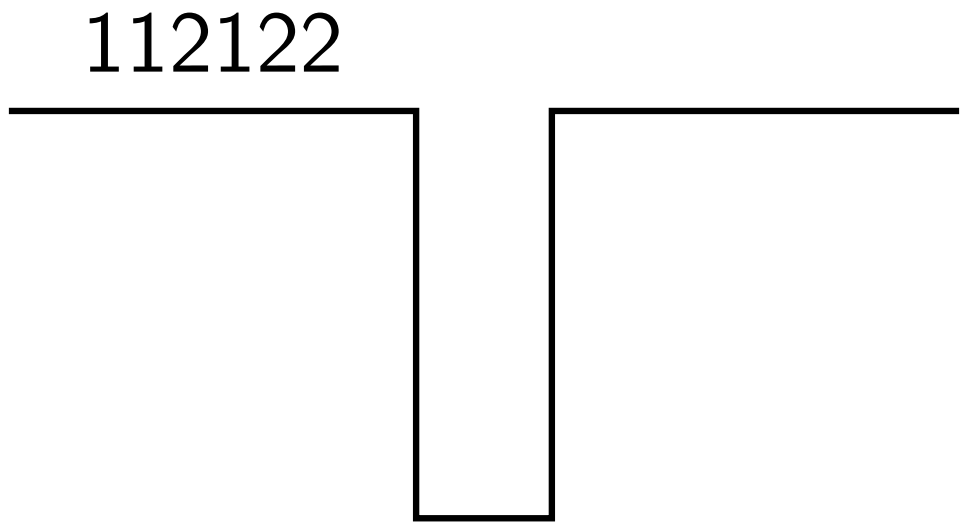


s_m allows m identical penguins in hole. s_2:



s_7 "more powerful" than s_6: every s_6-sortable word s_7-sortable.

s_m allows m identical penguins in hole. s_2:



s_7 "more powerful" than s_6: every s_6-sortable word s_7-sortable. But:

Theorem (L. 2026)

For all $m \neq m'$, there are words w such that w takes much longer to sort with s_m than $s_{m'}$.